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Syllabus: Combinatorics, Counting, Recurrence relations, Generating functions.

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	1	0	1	0	0	0	1	1	1	0	0
2 Marks Count	0	0	0	0	0	0	1	2	0	1	0
Total Marks	1	0	1	0	0	0	3	5	1	2	0

Welcome to the "Discrete Mathematics: Combinatory" chapter! This section is a cornerstone of computer science, providing the fundamental tools to count, arrange, and select objects, analyze algorithms, and understand probability. For the GATE Computer Science exam, combinatorics is crucial for topics like algorithm analysis (counting operations, permutations), data structures (hashing, tree traversals), probability, and even network protocols. Typically, you can expect 3-5 marks from this area, often in conjunction with probability or algorithm analysis. Questions range from direct application of formulas to more complex problem-solving scenarios requiring a combination of principles, sometimes involving tricky wording or subtle conditions like 'distinct' vs. 'identical' items, or 'empty' vs. 'non-empty' containers.

Topic-wise Key Concepts

Balls In Bins

Definition and Core Idea: The "Balls In Bins" paradigm is a versatile model for various counting problems. It involves distributing a certain number of "balls" (items) into a certain number of "bins" (containers or categories), considering whether the balls are identical or distinct, and whether the bins are identical or distinct, and if bins can be empty or must contain at least one ball.

Important Formulas, Theorems, and Results:

1. Distinct Balls into Distinct Bins:

- Any number of balls per bin:** If there are n distinct balls and k distinct bins, each ball can go into any of the k bins. Total ways: k^n .
- At most one ball per bin (injective mapping):** This is equivalent to permutations. Total ways: $P(k, n) = \frac{k!}{(k-n)!}$ if $k \geq n$, and 0 otherwise.

2. Identical Balls into Distinct Bins:

- Any number of balls per bin (stars and bars):** If there are n identical balls and k distinct bins, this is equivalent to finding non-negative integer solutions to $x_1 + x_2 + \dots + x_k = n$. Total ways: $C(n+k-1, k-1) = C(n+k-1, n)$.
- At least one ball per bin (stars and bars with pre-allocation):** First, place one ball in each bin, leaving $n-k$ balls to distribute. Total ways: $C((n-k)+k-1, k-1) = C(n-1, k-1)$ if $n \geq k$, and 0 otherwise.

3. Distinct Balls into Identical Bins: This is related to Stirling Numbers of the Second Kind.

- Any number of balls per bin:** Sum of Stirling numbers of the second kind. Total ways: $\sum_{i=1}^k S(n, i)$.
- At least one ball per bin:** Total ways: $S(n, k)$.

$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^j C(k, j) (k-j)^n$$

4. Identical Balls into Identical Bins: This is related to integer partitions.

- Any number of balls per bin:** The number of partitions of n into at most k parts, denoted $P(n, k)$.
- At least one ball per bin:** The number of partitions of n into exactly k parts, denoted $P_k(n)$. There is no simple closed-form formula; often solved using generating functions or recursion.

Key Properties and Identities:

- The "stars and bars" formula $C(n+k-1, k-1)$ is fundamental for problems involving distributing identical items into distinct categories or finding non-negative integer solutions.
- Understanding the distinction between "distinct" and "identical" for both balls and bins is paramount.

Common Pitfalls or Tricky Points:

- **Confusing Identical vs. Distinct:** This is the most common error. Always identify if the items being distributed and the containers are distinguishable or not.
- **Empty vs. Non-empty Bins:** Pay close attention to whether bins are allowed to be empty or if they must contain at least one item. This changes the formula significantly.
- **Overlapping Cases:** Ensure that cases are mutually exclusive and exhaustive when breaking down a problem.

Standard Problem-Solving Techniques or Shortcuts:

- **Mapping to Stars and Bars:** Many problems can be rephrased as distributing identical items into distinct bins. For example, finding the number of non-negative integer solutions to an equation.
- **Inclusion-Exclusion Principle:** For problems where "at least one" condition is difficult to apply directly, use inclusion-exclusion to subtract cases where bins are empty.
- **Complementary Counting:** Sometimes it's easier to count the total ways and subtract the "unwanted" ways.

Combinatory

Definition and Core Idea: Combinatorics is the branch of discrete mathematics concerned with counting, arrangement, and selection of objects. It deals with permutations (arrangements where order matters) and combinations (selections where order does not matter), often involving constraints or repetitions.

Important Formulas, Theorems, and Results:

1. Permutations (Order Matters):

- **Permutations of n distinct items taken k at a time:** $P(n, k) = \frac{n!}{(n-k)!}$.
- **Permutations of n distinct items taken n at a time:** $P(n, n) = n!$.
- **Permutations with Repetition (multiset permutations):** For n items where there are n_1 identical items of type 1, n_2 of type 2, ..., n_k of type k , and $n_1 + n_2 + \dots + n_k = n$. Total ways: $\frac{n!}{n_1!n_2!\dots n_k!}$.
- **Circular Permutations:** For n distinct items arranged in a circle, total ways: $(n-1)!$. If arrangements are symmetric (e.g., beads on a necklace), divide by 2: $\frac{(n-1)!}{2}$.

2. Combinations (Order Does Not Matter):

- **Combinations of n distinct items taken k at a time:** $C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$.
- **Combinations with Repetition (multiset combinations):** Selecting k items from n types with repetition allowed. Total ways: $C(n+k-1, k)$. This is equivalent to the "identical balls into distinct bins" problem.

3. Binomial Theorem:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

4. Multinomial Theorem:

$$(x_1 + x_2 + \dots + x_m)^n = \sum_{n_1 + \dots + n_m = n} \frac{n!}{n_1!n_2!\dots n_m!} x_1^{n_1} x_2^{n_2} \dots x_m^{n_m}$$

Key Properties and Identities:

- **Symmetry Identity:** $C(n, k) = C(n, n-k)$
- **Pascal's Identity:** $C(n, k) = C(n-1, k) + C(n-1, k-1)$
- **Sum of Binomial Coefficients:** $\sum_{k=0}^n C(n, k) = 2^n$
- **Hockey-stick Identity:** $\sum_{i=r}^n C(i, r) = C(n+1, r+1)$

Common Pitfalls or Tricky Points:

- **Overcounting/Undercounting:** Ensure each valid arrangement/selection is counted exactly once. This often happens when dealing with symmetry or identical items.
- **"At Least"/"At Most" Conditions:** These often require complementary counting or casework.
- **Distinguishing Permutations vs. Combinations:** If the order of selection or arrangement matters, it's a permutation. If not, it's a combination.

Standard Problem-Solving Techniques or Shortcuts:

- **Principle of Multiplication:** If a task can be broken into a sequence of k independent steps, and step i can be performed in n_i ways, then the total number of ways is $n_1 \times n_2 \times \dots \times n_k$.
- **Principle of Addition:** If a task can be performed in one of k mutually exclusive ways, and way i can be performed in n_i ways, then the total number of ways is $n_1 + n_2 + \dots + n_k$.
- **Complementary Counting:** Count the total number of ways and subtract the number of "unwanted" ways. Useful for "at least one" type problems.
- **Casework:** Break down a complex problem into simpler, mutually exclusive cases and sum the results.

Counting

Definition and Core Idea: Counting, in a broader sense, encompasses the fundamental principles used to determine the size of finite sets. It builds upon combinatorial concepts and introduces powerful tools like the Principle of Inclusion-Exclusion to handle complex scenarios involving overlapping sets.

Important Formulas, Theorems, and Results:

1. Principle of Inclusion-Exclusion (PIE):

- For two sets: $|A \cup B| = |A| + |B| - |A \cap B|$
- For three sets: $|A \cup B \cup C| = |A| + |B| + |C| - (|A \cap B| + |A \cap C| + |B \cap C|) + |A \cap B \cap C|$
- General form for n sets: $|\bigcup_{i=1}^n A_i| = \sum |A_i| - \sum |A_i \cap A_j| + \dots + (-1)^{n+1} |A_1 \cap \dots \cap A_n|$

2. Derangements: A permutation of n objects such that no object remains in its original position. The number of derangements of n objects, denoted D_n or $!n$.

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!} = n! \left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \dots + \frac{(-1)^n}{n!} \right)$$

For large n , $D_n \approx \frac{n!}{e}$.

Key Properties and Identities:

- PIE is essential for counting elements in the union of non-disjoint sets.
- Derangements are a specific application of PIE for permutations with no fixed points.

Common Pitfalls or Tricky Points:

- **Incorrectly Identifying Sets for PIE:** Clearly define the properties that form the sets $A, B, C, \dots$ and their intersections.
- **Misapplying PIE:** Ensure you are alternating signs correctly and covering all necessary intersections.
- **"At Least One" vs. "Exactly One":** PIE typically helps with "at least one" conditions. For "exactly one," you might need to combine PIE with complementary counting or other techniques.

Standard Problem-Solving Techniques or Shortcuts:

- **Casework:** Break down complex problems into simpler, mutually exclusive scenarios.
- **Complementary Counting:** Count the total possibilities and subtract the undesirable ones. Often used in conjunction with PIE.
- **Direct Application of PIE:** For problems involving properties that overlap, PIE is usually the most direct approach.

Generating Functions

Definition and Core Idea: A generating function is a formal power series where the coefficients represent the terms of a sequence. It provides a powerful algebraic tool for solving recurrence relations, counting problems, and proving combinatorial identities by converting sequence operations into algebraic manipulations.

Important Formulas, Theorems, and Results:

1. **Definition:** For a sequence $a_0, a_1, a_2, \dots$, its ordinary generating function (OGF) is $G(x) = \sum_{n=0}^{\infty} a_n x^n$.
2. **Common Generating Functions:**

- **Constant sequence** $1, 1, 1, \dots: G(x) = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$.
 - **Arithmetic progression** $1, a, a^2, \dots: G(x) = \sum_{n=0}^{\infty} (ax)^n = \frac{1}{1-ax}$.
 - **Binomial coefficients** $C(n, k)$ for fixed $n: G(x) = \sum_{k=0}^n C(n, k)x^k = (1+x)^n$.
 - **Combinations with repetition** $C(n+k-1, k)$ for fixed $n: G(x) = \sum_{k=0}^{\infty} C(n+k-1, k)x^k = \frac{1}{(1-x)^n}$.
 - **Sequence** $a_n = n: G(x) = \sum_{n=0}^{\infty} nx^n = \frac{x}{(1-x)^2}$.
 - **Sequence** $a_n = n^2: G(x) = \sum_{n=0}^{\infty} n^2x^n = \frac{x(1+x)}{(1-x)^3}$.
3. **Operations on Generating Functions:**
- **Addition:** If $A(x) = \sum a_n x^n$ and $B(x) = \sum b_n x^n$, then $A(x) + B(x) = \sum (a_n + b_n)x^n$.
 - **Multiplication (Convolution):** If $A(x) = \sum a_n x^n$ and $B(x) = \sum b_n x^n$, then $A(x)B(x) = \sum c_n x^n$, where $c_n = \sum_{k=0}^n a_k b_{n-k}$. This is crucial for counting combinations of independent choices.
 - **Differentiation:** $G'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$. So, $xG'(x) = \sum_{n=1}^{\infty} n a_n x^n$.
 - **Integration:** $\int G(x) dx = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$.
 - **Shift:** $x^k G(x) = \sum_{n=0}^{\infty} a_n x^{n+k}$ (shifts coefficients to the right). $\frac{G(x) - a_0 - \dots - a_{k-1}x^{k-1}}{x^k} = \sum_{n=k}^{\infty} a_n x^{n-k}$ (shifts coefficients to the left).

Key Properties and Identities:

- The coefficient of x^n in a generating function $G(x)$ is a_n , denoted $[x^n]G(x)$.
- Generating functions are particularly useful for problems involving sums, combinations with repetition, and solving linear recurrence relations.

Common Pitfalls or Tricky Points:

- **Forming the Correct GF:** Carefully translate the problem's constraints (e.g., minimum/maximum number of items, distinct/identical items) into the terms of the generating function.
- **Extracting Coefficients:** This often requires partial fraction decomposition for rational functions or using generalized binomial theorem.
- **Confusion with Exponential Generating Functions (EGFs):** EGFs are used for permutations, while OGFs are for combinations. GATE typically focuses on OGFs.

Standard Problem-Solving Techniques or Shortcuts:

- **Solving Recurrence Relations:**
 1. Write the recurrence relation.
 2. Multiply by x^n and sum over all valid n .
 3. Manipulate the sums to relate them to the generating function $G(x)$.
 4. Solve for $G(x)$ as a rational function.
 5. Use partial fraction decomposition and known series expansions to find the coefficient of x^n .
- **Counting Combinations:** Represent each type of item or choice as a factor in the generating function. The coefficient of x^n in the product of these factors gives the number of ways to choose n items.

Modular Arithmetic

Definition and Core Idea: Modular arithmetic is a system of arithmetic for integers, where numbers "wrap around" when they reach a certain value—the modulus. It's fundamental in cryptography, number theory, and hashing, and is used in GATE for problems involving remainders, large powers, and divisibility.

Important Formulas, Theorems, and Results:

1. **Congruence Relation:** $a \equiv b \pmod{m}$ if m divides $(a - b)$, or equivalently, a and b have the same remainder when divided by m .
2. **Basic Properties:**
 - If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then:
 - $a + c \equiv b + d \pmod{m}$
 - $a - c \equiv b - d \pmod{m}$
 - $ac \equiv bd \pmod{m}$

- $a^k \equiv b^k \pmod{m}$ for any positive integer k .
- 3. **Modular Inverse:** An integer a^{-1} is the modular multiplicative inverse of a modulo m if $aa^{-1} \equiv 1 \pmod{m}$.
 - An inverse exists if and only if $\gcd(a, m) = 1$.
 - Can be found using the Extended Euclidean Algorithm.
- 4. **Euler's Totient Function $\phi(n)$:** Counts the number of positive integers up to n that are relatively prime to n .
 - If $n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$ (prime factorization), then $\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_r}\right)$.
 - If p is prime, $\phi(p) = p - 1$.
 - If p is prime, $\phi(p^k) = p^k - p^{k-1}$.
- 5. **Fermat's Little Theorem:** If p is a prime number, then for any integer a not divisible by p , $a^{p-1} \equiv 1 \pmod{p}$. Also, $a^p \equiv a \pmod{p}$ for any integer a .
- 6. **Euler's Theorem:** If n and a are coprime positive integers, then $a^{\phi(n)} \equiv 1 \pmod{n}$. (Fermat's Little Theorem is a special case of Euler's Theorem when n is prime).
- 7. **Chinese Remainder Theorem (CRT):** (Conceptually important, full formula rarely needed for GATE) If integers $n_1, n_2, \dots, n_k$ are pairwise coprime, then for any given integers $a_1, a_2, \dots, a_k$, there exists an integer x that solves the system of congruences:

$$x \equiv a_1 \pmod{n_1}$$

$$x \equiv a_2 \pmod{n_2}$$

⋮

$$x \equiv a_k \pmod{n_k}$$

Moreover, all solutions x are congruent modulo $N = n_1 n_2 \dots n_k$.

Key Properties and Identities:

- Modular arithmetic allows us to work with remainders, simplifying calculations involving very large numbers.
- The concept of modular inverse is critical for division in modular arithmetic (multiplication by inverse).
- Fermat's Little Theorem and Euler's Theorem are powerful for simplifying large exponents modulo a number.

Common Pitfalls or Tricky Points:

- **Division:** Division is not directly defined. Instead, multiply by the modular inverse. If the inverse doesn't exist (i.e., $\gcd(a, m) \neq 1$), division is not possible.
- **Negative Results:** If a calculation yields a negative number, add the modulus until it's positive. E.g., $-5 \equiv 2 \pmod{7}$.
- **Large Exponents:** Always reduce the exponent modulo $\phi(m)$ (if $\gcd(a, m) = 1$) or modulo $m - 1$ (if m is prime and $\gcd(a, m) = 1$) to simplify calculations using Euler's or Fermat's theorem.

Standard Problem-Solving Techniques or Shortcuts:

- **Repeated Squaring (Exponentiation by Squaring):** Efficiently compute $a^b \pmod{m}$ by repeatedly squaring the base and reducing modulo m .
Example: $a^b = a^{b/2} \cdot a^{b/2}$ if b is even, $a^b = a \cdot a^{(b-1)/2} \cdot a^{(b-1)/2}$ if b is odd.
- **Finding Modular Inverse:**
 - **Extended Euclidean Algorithm:** For $ax + my = \gcd(a, m)$. If $\gcd(a, m) = 1$, then $ax \equiv 1 \pmod{m}$, so x is the inverse.
 - **Fermat's Little Theorem:** If m is prime, $a^{m-2} \equiv a^{-1} \pmod{m}$.
 - **Euler's Theorem:** If $\gcd(a, m) = 1$, $a^{\phi(m)-1} \equiv a^{-1} \pmod{m}$.
- **Properties of Congruence:** Apply addition, subtraction, multiplication properties to simplify expressions before taking the final modulo.

Pigeonhole Principle

Definition and Core Idea: The Pigeonhole Principle (PHP) is a fundamental concept stating that if n items are put into

m containers, with $n > m$, then at least one container must contain more than one item. It's a simple yet powerful tool for proving existence in combinatorial problems.

Important Formulas, Theorems, and Results:

- Basic Pigeonhole Principle:** If n pigeons are placed into m pigeonholes, and $n > m$, then at least one pigeonhole contains more than one pigeon.
- Generalized Pigeonhole Principle:** If n pigeons are placed into m pigeonholes, then at least one pigeonhole must contain at least $\lceil n/m \rceil$ pigeons.

Key Properties and Identities:

- PHP is an existence principle; it tells you that something must happen, but not how to find it or how many times it happens.
- It's often used in proofs to show that a certain condition must hold true.

Common Pitfalls or Tricky Points:

- Identifying Pigeons and Pigeonholes:** The most crucial step is correctly defining what constitutes a "pigeon" and a "pigeonhole" in the context of the problem. This often requires creative thinking.
- Applying Generalized PHP:** Ensure you correctly calculate $\lceil n/m \rceil$.
- "At Least" vs. "Exactly":** PHP guarantees "at least" a certain number; it doesn't guarantee "exactly."

Standard Problem-Solving Techniques or Shortcuts:

- Creative Definitions:** The key is to define pigeons and pigeonholes in a way that makes the principle applicable.
 - Pigeons are often the items being distributed or selected.
 - Pigeonholes are often categories, properties, or possible outcomes.
- Proof by Contradiction:** Sometimes, assuming the opposite of what you want to prove (i.e., no pigeonhole has more than k pigeons) can lead to a contradiction using PHP.
- Small Examples:** Test with small numbers to understand the problem structure and how to define pigeons/pigeonholes.

Recurrence Relation

Definition and Core Idea: A recurrence relation is an equation that defines a sequence where each term is given as a function of its preceding terms. They are fundamental in algorithm analysis (e.g., complexity of recursive algorithms), combinatorics (e.g., Fibonacci numbers), and discrete probability.

Important Formulas, Theorems, and Results:

- Linear Homogeneous Recurrence Relations with Constant Coefficients:**

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

- Characteristic Equation:** $r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_k = 0$
- Distinct Roots:** If $r_1, r_2, \dots, r_k$ are distinct roots, the general solution is $a_n = A_1 r_1^n + A_2 r_2^n + \dots + A_k r_k^n$.
- Repeated Roots:** If a root r has multiplicity m , its contribution to the general solution is $(A_1 + A_2 n + \dots + A_m n^{m-1}) r^n$.
- The constants A_i are determined by the initial conditions.

- Linear Non-Homogeneous Recurrence Relations:**

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + F(n)$$

- The general solution is $a_n = a_n^{(h)} + a_n^{(p)}$, where $a_n^{(h)}$ is the homogeneous solution and $a_n^{(p)}$ is a particular solution.
- Finding $a_n^{(p)}$:** Guess a form for $a_n^{(p)}$ based on $F(n)$.
 - If $F(n)$ is a polynomial of degree d , guess $P(n) = C_d n^d + \dots + C_0$. If 1 is a root of the characteristic equation with multiplicity s , multiply by n^s .
 - If $F(n)$ is an exponential $C \cdot b^n$, guess $A \cdot b^n$. If b is a root of the characteristic equation with multiplicity s ,

multiply by n^8 .

3. **Master Theorem (for Divide and Conquer Recurrences):** For recurrences of the form $T(n) = aT(n/b) + f(n)$ where $a > 1$, $b > 1$ and $f(n)$ is asymptotically positive.

1. If $f(n) = O(n^{\log_b a - \epsilon})$ for some $\epsilon > 0$, then $T(n) = \Theta(n^{\log_b a})$.
2. If $f(n) = \Theta(n^{\log_b a})$, then $T(n) = \Theta(n^{\log_b a} \log n)$.
3. If $f(n) = \Omega(n^{\log_b a + \epsilon})$ for some $\epsilon > 0$, AND $af(n/b) \leq cf(n)$ for some $c < 1$ and sufficiently large n (regularity condition), then $T(n) = \Theta(f(n))$.

Key Properties and Identities:

- The order of a recurrence relation is the difference between the largest and smallest indices.
- Linear recurrence relations with constant coefficients are solvable using the characteristic equation method.
- The Master Theorem provides a quick way to find asymptotic bounds for many divide-and-conquer algorithms.

Common Pitfalls or Tricky Points:

- **Incorrect Characteristic Equation:** Ensure all terms are on one side and coefficients are correct.
- **Handling Repeated Roots:** Don't forget the $n, n^2, \dots$ factors for repeated roots.
- **Finding Particular Solutions:** The guess for $a_n^{(p)}$ must be adjusted if it overlaps with the homogeneous solution.
- **Boundary Conditions:** Always use initial conditions to find the constants A_i in the general solution.
- **Master Theorem Conditions:** Carefully check all three cases and the regularity condition for case 3.

Standard Problem-Solving Techniques or Shortcuts:

- **Characteristic Equation Method:** The standard approach for linear homogeneous recurrences.
- **Substitution Method:** For simpler recurrences, guess a solution and prove it by induction.
- **Generating Functions:** A powerful general method for solving linear recurrences (see Generating Functions section).
- **Master Theorem:** For divide-and-conquer algorithms, this is the fastest way to find the asymptotic complexity.

Summation

Definition and Core Idea: Summation involves finding the sum of a sequence of numbers, often expressed using sigma notation. It's crucial for analyzing algorithm complexity, calculating probabilities, and working with series in various mathematical contexts.

Important Formulas, Theorems, and Results:

1. **Arithmetic Series:** Sum of an arithmetic progression $a, a + d, a + 2d, \dots, a + (n - 1)d$.

$$\sum_{i=0}^{n-1} (a + id) = \frac{n}{2}(2a + (n - 1)d)$$

If $a = 1, d = 1$, sum of first n integers: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.

2. **Geometric Series:** Sum of a geometric progression $a, ar, ar^2, \dots, ar^{n-1}$.

$$\sum_{i=0}^{n-1} ar^i = a \frac{1 - r^n}{1 - r} \quad (\text{if } r \neq 1)$$

Infinite Geometric Series: If $|r| < 1$, then $\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r}$.

3. **Sum of Squares:**

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

4. **Sum of Cubes:**

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2$$

5. **Harmonic Series:** $H_n = \sum_{i=1}^n \frac{1}{i}$. $H_n \approx \ln n + \gamma$, where $\gamma \approx 0.577$ (Euler-Mascheroni constant).
 6. **Telescoping Sums:** A sum where intermediate terms cancel out.

$$\sum_{i=1}^n (a_i - a_{i+1}) = (a_1 - a_2) + (a_2 - a_3) + \cdots + (a_n - a_{n+1}) = a_1 - a_{n+1}$$

Key Properties and Identities:

- Linearity:** $\sum (c \cdot a_i + d \cdot b_i) = c \sum a_i + d \sum b_i$.
- Changing the limits of summation requires careful adjustment of the terms.

Common Pitfalls or Tricky Points:

- Off-by-One Errors:** Incorrectly determining the number of terms in a series (e.g., summing from $i = 0$ to $n - 1$ vs. $i = 1$ to n).
- Incorrectly Identifying Series Type:** Make sure it's truly arithmetic, geometric, or another standard type before applying formulas.
- Convergence of Infinite Series:** For infinite geometric series, ensure $|r| < 1$.

Standard Problem-Solving Techniques or Shortcuts:

- Direct Formula Application:** For standard series (arithmetic, geometric, sums of powers), memorize and apply the formulas.
- Telescoping:** Look for terms that can be expressed as differences, leading to cancellation.
- Splitting Sums:** Break a complex sum into multiple simpler sums using linearity.
- Changing Index:** Adjust the summation index to match standard formulas (e.g., starting from 0 or 1).
- Integral Approximation:** For sums that don't have a closed form, integrals can provide good upper and lower bounds (e.g., for harmonic series).

Quick Formula Reference

- Permutations:**
 - $P(n, k) = \frac{n!}{(n-k)!}$
 - $P(n, n) = n!$
 - Permutations with Repetition: $\frac{n!}{n_1! n_2! \dots n_k!}$
 - Circular Permutations: $(n-1)!$
- Combinations:**
 - $C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$
 - Combinations with Repetition: $C(n+k-1, k)$
 - Symmetry: $C(n, k) = C(n, n-k)$
 - Pascal's Identity: $C(n, k) = C(n-1, k) + C(n-1, k-1)$
 - Sum: $\sum_{k=0}^n C(n, k) = 2^n$
- Balls in Bins:**
 - Distinct Balls, Distinct Bins (any per bin): k^n
 - Distinct Balls, Distinct Bins (at most one per bin): $P(k, n)$
 - Identical Balls, Distinct Bins (any per bin): $C(n+k-1, k-1)$
 - Identical Balls, Distinct Bins (at least one per bin): $C(n-1, k-1)$
 - Distinct Balls, Identical Bins (at least one per bin): $S(n, k)$ (Stirling numbers of 2nd kind)
- Principle of Inclusion-Exclusion (PIE):**
 - $|A \cup B| = |A| + |B| - |A \cap B|$
 - $|A \cup B \cup C| = |A| + |B| + |C| - (|A \cap B| + |A \cap C| + |B \cap C|) + |A \cap B \cap C|$
- Derangements:** $D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!} \approx \frac{n!}{e}$
- Generating Functions:**

$$\begin{aligned}
 & \circ \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \\
 & \circ \frac{1}{1-ax} = \sum_{n=0}^{\infty} (ax)^n \\
 & \circ (1+x)^n = \sum_{k=0}^n C(n, k)x^k \\
 & \circ \frac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)x^k \\
 & \circ \frac{x}{(1-x)^2} = \sum_{n=0}^{\infty} nx^n
 \end{aligned}$$

• Modular Arithmetic:

- Fermat's Little Theorem (p prime, $\gcd(a, p) = 1$): $a^{p-1} \equiv 1 \pmod{p}$
- Euler's Theorem ($\gcd(a, n) = 1$): $a^{\phi(n)} \equiv 1 \pmod{n}$
- Euler's Totient Function: $\phi(n) = n \prod_{p|n, p \text{ prime}} (1 - \frac{1}{p})$

• Pigeonhole Principle:

- Basic: If n pigeons in m holes and $n > m$, at least one hole has >1 pigeon.
- Generalized: At least one hole has $\lceil n/m \rceil$ pigeons.

• Recurrence Relations:

- Homogeneous Linear: $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$ (solve characteristic equation $r^k - c_1 r^{k-1} - \dots - c_k = 0$).
- Master Theorem: For $T(n) = aT(n/b) + f(n)$. Compare $f(n)$ with $n^{\log_b a}$.

• Summation Formulas:

- Arithmetic Series: $\sum_{i=0}^{n-1} (a + id) = \frac{n}{2} (2a + (n-1)d)$
- Sum of first n integers: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$
- Geometric Series: $\sum_{i=0}^{n-1} ar^i = a \frac{1-r^n}{1-r}$
- Infinite Geometric Series ($|r| < 1$): $\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r}$
- Sum of Squares: $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$
- Sum of Cubes: $\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2$

Important Tips for GATE

1. **Read Carefully:** Pay extreme attention to keywords like "distinct," "identical," "at least," "at most," "with replacement," "without replacement," "ordered," "unordered," "empty bins allowed," etc. These words dictate which formula or technique to use.
2. **Visualize and Simplify:** For combinatorics and PHP problems, try to draw small examples or simplify the problem to a smaller scale to understand the underlying structure. This helps in correctly identifying pigeons/pigeonholes or the counting method.
3. **Master the Basics:** Ensure you are fluent with fundamental permutations, combinations, and the stars and bars technique. Many complex problems are built upon these basic principles.
4. **Practice with PYQs:** Solve previous year GATE questions diligently. This helps you understand the common patterns, tricky phrasing, and expected depth of knowledge. Pay attention to how different concepts are combined in a single question.
5. **Complementary Counting:** For "at least" or "not allowed" type problems, consider using complementary counting (Total ways - Unwanted ways). This can often simplify the problem significantly compared to direct counting.
6. **Time Management for Recurrence Relations:** For recurrence relations, if it's a divide-and-conquer type, try the Master Theorem first. If it's a linear recurrence, be quick with characteristic equations and solving for constants using initial conditions.
7. **Modular Arithmetic for Large Numbers:** Always reduce intermediate results modulo m in modular arithmetic. For large exponents, use Fermat's Little Theorem or Euler's Theorem to reduce the exponent modulo $\phi(m)$ (or $p-1$ if m is prime).
8. **Don't Be Afraid of Casework:** If a problem seems too complex for a single formula, break it down into mutually exclusive cases. Calculate each case separately and sum the results. Ensure cases are exhaustive.

1.1

Balls In Bins (6)

1.1.1 Balls In Bins: GATE CSE 1999 | Question: 1.3



The number of binary strings of n zeros and k ones in which no two ones are adjacent is

- A. ${}^{n-1}C_k$ B. nC_k C. ${}^nC_{k+1}$ D. None of the above

Answer key

1.1.2 Balls In Bins: GATE CSE 2000 | Question: 5



A multiset is an unordered collection of elements where elements may repeat any number of times. The size of a multiset is the number of elements in it, counting repetitions.

- What is the number of multisets of size 4 that can be constructed from n distinct elements so that at least one element occurs exactly twice?
- How many multisets can be constructed from n distinct elements?

gatecse-2000 combinatorial normal descriptive counting balls-in-bins

Answer key

1.1.3 Balls In Bins: GATE CSE 2002 | Question: 13



- In how many ways can a given positive integer $n > 2$ be expressed as the sum of 2 positive integers (which are not necessarily distinct). For example, for $n = 3$, the number of ways is 2, i.e., $1 + 2, 2 + 1$. Give only the answer without any explanation.
- In how many ways can a given positive integer $n > 3$ be expressed as the sum of 3 positive integers (which are not necessarily distinct). For example, for $n = 4$, the number of ways is 3, i.e., $1 + 2 + 1, 2 + 1 + 1$ and $1 + 1 + 2$. Give only the answer without explanation.
- In how many ways can a given positive integer $n > k$ be expressed as the sum of k positive integers (which are not necessarily distinct). Give only the answer without explanation.

gatecse-2002 combinatorial normal descriptive balls-in-bins

Answer key

1.1.4 Balls In Bins: GATE CSE 2003 | Question: 34



m identical balls are to be placed in n distinct bags. You are given that $m > kn$, where k is a natural number > 1 . In how many ways can the balls be placed in the bags if each bag must contain at least k balls?

- $\binom{m-k}{n-1}$
- $\binom{m-kn+n-1}{n-1}$
- $\binom{m-1}{n-k}$
- $\binom{m-kn+n+k-2}{n-k}$

gatecse-2003 combinatorial balls-in-bins normal

Answer key

1.1.5 Balls In Bins: GATE CSE 2022 | Question: 22



The number of arrangements of six identical balls in three identical bins is _____.

gatecse-2022 numerical-answers combinatorial balls-in-bins one-mark

Answer key

1.1.6 Balls In Bins: GATE IT 2004 | Question: 35



In how many ways can we distribute 5 distinct balls, $B_1, B_2, \dots, B_5$ in 5 distinct cells, $C_1, C_2, \dots, C_5$ such that Ball B_i is not in cell $C_i, \forall i = 1, 2, \dots, 5$ and each cell contains exactly one ball?

- 44
- 96
- 120
- 3125

gateit-2004 combinatorial normal balls-in-bins

Answer key

1.2.1 Combinatory: GATE CSE 1990 | Question: 3-ix



The number of ways in which 5 A 's, 5 B 's and 5 C 's can be arranged in a row is:

- A. $15!/(5!)^3$ B. $15!$ C. $\binom{15}{5}$ D. $15!(5!3!)$

gate1990 normal combinatory

Answer key

1.2.2 Combinatory: GATE CSE 1991 | Question: 16,a



Find the number of binary strings w of length $2n$ with an equal number of 1's and 0's and the property that every prefix of w has at least as many 0's as 1's.

gate1991 combinatory normal descriptive catalan-number

Answer key

1.2.3 Combinatory: GATE CSE 1998 | Question: 1.23



How many substrings of different lengths (non-zero) can be formed from a character string of length n ?

- A. n B. n^2 C. 2^n D. $\frac{n(n+1)}{2}$

gate1998 combinatory normal

Answer key

1.2.4 Combinatory: GATE CSE 1999 | Question: 2.2



Two girls have picked 10 roses, 15 sunflowers and 15 daffodils. What is the number of ways they can divide the flowers among themselves?

- A. 1638 B. 2100 C. 2640 D. None of the above

gate1999 combinatory normal

Answer key

1.2.5 Combinatory: GATE CSE 2001 | Question: 2.1



How many 4-digit even numbers have all 4 digits distinct?

- A. 2240 B. 2296 C. 2620 D. 4536

gatecse-2001 combinatory normal

Answer key

1.2.6 Combinatory: GATE CSE 2003 | Question: 4



Let A be a sequence of 8 distinct integers sorted in ascending order. How many distinct pairs of sequences, B and C are there such that

- each is sorted in ascending order,
- B has 5 and C has 3 elements, and
- the result of merging B and C gives A

- A. 2 B. 30 C. 56 D. 256

gatecse-2003 combinatory normal

Answer key

1.2.7 Combinatory: GATE CSE 2003 | Question: 5



n couples are invited to a party with the condition that every husband should be accompanied by his wife. However, a wife need not be accompanied by her husband. The number of different gatherings possible at the party is

A. $2^n C_n \times 2^n$

B. 3^n

C. $\frac{(2n)!}{2^n}$

D. $2^n C_n$

gatecse-2003 combinatory normal

Answer key

1.2.8 Combinatory: GATE CSE 2004 | Question: 75



Mala has the colouring book in which each English letter is drawn two times. She wants to paint each of these 52 prints with one of k colours, such that the colour pairs used to colour any two letters are different. Both prints of a letter can also be coloured with the same colour. What is the minimum value of k that satisfies this requirement?

A. 9

B. 8

C. 7

D. 6

gatecse-2004 combinatory

Answer key

1.2.9 Combinatory: GATE CSE 2007 | Question: 84



Suppose that a robot is placed on the Cartesian plane. At each step it is allowed to move either one unit up or one unit right, i.e., if it is at (i, j) then it can move to either $(i + 1, j)$ or $(i, j + 1)$.

How many distinct paths are there for the robot to reach the point $(10, 10)$ starting from the initial position $(0, 0)$?

A. ${}^{20}C_{10}$

B. 2^{20}

C. 2^{10}

D. None of the above

gatecse-2007 combinatory

Answer key

1.2.10 Combinatory: GATE CSE 2007 | Question: 85



Suppose that a robot is placed on the Cartesian plane. At each step it is allowed to move either one unit up or one unit right, i.e., if it is at (i, j) then it can move to either $(i + 1, j)$ or $(i, j + 1)$.

Suppose that the robot is not allowed to traverse the line segment from $(4, 4)$ to $(5, 4)$. With this constraint, how many distinct paths are there for the robot to reach $(10, 10)$ starting from $(0, 0)$?

A. 2^9

C. ${}^8C_4 \times {}^{11}C_5$

B. 2^{19}

D. ${}^{20}C_{10} - {}^8C_4 \times {}^{11}C_5$

gatecse-2007 combinatory normal discrete-mathematics

Answer key

1.2.11 Combinatory: GATE CSE 2014 | Set 1 | Question: 49



A pennant is a sequence of numbers, each number being 1 or 2. An n -pennant is a sequence of numbers with sum equal to n . For example, $(1, 1, 2)$ is a 4-pennant. The set of all possible 1-pennants is (1) , the set of all possible 2-pennants is $(2), (1, 1)$ and the set of all 3-pennants is $(2, 1), (1, 1, 1), (1, 2)$. Note that the pennant $(1, 2)$ is not the same as the pennant $(2, 1)$. The number of 10-pennants is _____.

gatecse-2014-set1 combinatory numerical-answers normal

Answer key

1.2.12 Combinatory: GATE CSE 2015 | Set 3 | Question: 5



The number of 4 digit numbers having their digits in non-decreasing order (from left to right) constructed by using the digits belonging to the set $\{1, 2, 3\}$ is _____.

gatecse-2015-set3 combinatory normal numerical-answers

Answer key

1.2.13 Combinatory: GATE CSE 2019 | Question: 5



Let $U = \{1, 2, \dots, n\}$ Let $A = \{(x, X) \mid x \in X, X \subseteq U\}$. Consider the following two statements on $|A|$.

- I. $|A| = n2^{n-1}$
 II. $|A| = \sum_{k=1}^n k \binom{n}{k}$

Which of the above statements is/are TRUE?

- A. Only I B. Only II C. Both I and II D. Neither I nor II

gatecse-2019 engineering-mathematics discrete-mathematics combinatory one-mark

Answer key

1.2.14 Combinatory: GATE CSE 2020 | Question: 42



The number of permutations of the characters in LILAC so that no character appears in its original position, if the two L's are indistinguishable, is _____.

gatecse-2020 numerical-answers combinatory two-marks

Answer key

1.2.15 Combinatory: GATE CSE 2025 | Set 1 | Question: 20



Let S be the set of all ternary strings defined over the alphabet $\{a, b, c\}$. Consider all strings in S that contain at least one occurrence of two consecutive symbols, that is, "aa", "bb" or "cc". The number of such strings of length 5 that are possible is _____. (Answer in integer).

gatecse2025-set1 combinatory numerical-answers one-mark

Answer key

1.2.16 Combinatory: GATE CSE 2026 | Set 1 | Question: 2



Consider 4×4 matrices with their elements from $\{0, 1\}$. The number of such matrices with even number of 1s in every row and every column is _____.

- A. 512 B. 1025 C. 1023 D. 255

gatecse-2026-set1 combinatory one-mark

Answer key

1.2.17 Combinatory: GATE IT 2005 | Question: 46



A line L in a circuit is said to have a *stuck-at-0* fault if the line permanently has a logic value 0. Similarly a line L in a circuit is said to have a *stuck-at-1* fault if the line permanently has a logic value 1. A circuit is said to have a multiple *stuck-at* fault if one or more lines have stuck at faults. The total number of distinct multiple *stuck-at* faults possible in a circuit with N lines is _____.

- A. 3^N B. $3^N - 1$ C. $2^N - 1$ D. 2

gateit-2005 combinatory normal

Answer key

1.2.18 Combinatory: GATE IT 2008 | Question: 25



In how many ways can b blue balls and r red balls be distributed in n distinct boxes?

- A. $\frac{(n+b-1)!(n+r-1)!}{(n-1)!b!(n-1)!r!}$
 B. $\frac{(n+(b+r)-1)!}{(n-1)!(n-1)!(b+r)!}$
 C. $\frac{n!}{b!r!}$

D. $\frac{(n + (b + r) - 1)!}{n! (b + r - 1)}$

gateit-2008 combinatory normal

Answer key

1.3 Counting (6)

1.3.1 Counting: GATE CSE 1990 | Question: 3-iii

The number of rooted binary trees with n nodes is,

- A. Equal to the number of ways of multiplying $(n + 1)$ matrices.
 B. Equal to the number of ways of arranging n out of $2n$ distinct elements.
 C. Equal to $\frac{1}{(n+1)} \binom{2n}{n}$.
 D. Equal to $n!$

gate1990 normal combinatory catalan-number multiple-selects counting

Answer key

1.3.2 Counting: GATE CSE 1991 | Question: 02-iv

Match the pairs in the following questions by writing the corresponding letters only.

A.	The number of distinct binary tree with n nodes.	P.	$\frac{n!}{2}$
B.	The number of binary strings of the length of $2n$ with an equal number of 0's and 1's	Q.	$\binom{3n}{n}$
C.	The number of even permutation of n objects.	R.	$\binom{2n}{n}$
D.	The number of binary strings of length $6n$ which are palindromes with $2n$ 0's.	S.	$\frac{1}{1+n} \binom{2n}{n}$

gate1991 combinatory normal match-the-following counting

Answer key

1.3.3 Counting: GATE CSE 1994 | Question: 1.15

The number of substrings (of all lengths inclusive) that can be formed from a character string of length n is

- A. n B. n^2 C. $\frac{n(n-1)}{2}$ D. $\frac{n(n+1)}{2}$

gate1994 combinatory counting normal

Answer key

1.3.4 Counting: GATE CSE 2021 | Set 1 | Question: 19

There are 6 jobs with distinct difficulty levels, and 3 computers with distinct processing speeds. Each job is assigned to a computer such that:

- The fastest computer gets the toughest job and the slowest computer gets the easiest job.
- Every computer gets at least one job.

The number of ways in which this can be done is _____.

gatecse-2021-set1 combinatory counting numerical-answers one-mark

Answer key

1.3.5 Counting: GATE CSE 2021 | Set 2 | Question: 50



Let S be a set consisting of 10 elements. The number of tuples of the form (A, B) such that A and B are subsets of S , and $A \subset B$ is _____.

gatecse-2021-set2 combinatorial counting numerical-answers two-marks

Answer key

1.3.6 Counting: GATE CSE 2023 | Question: 38



Let $U = \{1, 2, \dots, n\}$, where n is a large positive integer greater than 1000. Let k be a positive integer less than n . Let A, B be subsets of U with $|A| = |B| = k$ and $A \cap B = \emptyset$. We say that a permutation of U separates A from B if one of the following is true.

- All members of A appear in the permutation before any of the members of B .
- All members of B appear in the permutation before any of the members of A .

How many permutations of U separate A from B ?

A. $n!$

C. $\binom{n}{2k} (n - 2k)! (k!)^2$

B. $\binom{n}{2k} (n - 2k)!$
D. $2 \binom{n}{2k} (n - 2k)! (k!)^2$

gatecse-2023 combinatorial counting two-marks

Answer key

1.4 Generating Functions (6)

1.4.1 Generating Functions: GATE CSE 1987 | Question: 10b



What is the generating function $G(z)$ for the sequence of Fibonacci numbers?

gate1987 combinatorial generating-functions descriptive

Answer key

1.4.2 Generating Functions: GATE CSE 2005 | Question: 50



Let $G(x) = \frac{1}{(1-x)^2} = \sum_{i=0}^{\infty} g(i)x^i$, where $|x| < 1$. What is $g(i)$?

A. i

B. $i + 1$

C. $2i$

D. 2^i

gatecse-2005 normal generating-functions

Answer key

1.4.3 Generating Functions: GATE CSE 2016 | Set 1 | Question: 26



The coefficient of x^{12} in $(x^3 + x^4 + x^5 + x^6 + \dots)^3$ is _____.

gatecse-2016-set1 combinatorial generating-functions normal numerical-answers

Answer key

1.4.4 Generating Functions: GATE CSE 2017 | Set 2 | Question: 47



If the ordinary generating function of a sequence $\{a_n\}_{n=0}^{\infty}$ is $\frac{1+z}{(1-z)^3}$, then $a_3 - a_0$ is equal to _____.

gatecse-2017-set2 combinatorial generating-functions numerical-answers normal

Answer key

1.4.5 Generating Functions: GATE CSE 2018 | Question: 1



Which one of the following is a closed form expression for the generating function of the sequence $\{a_n\}$, where $a_n = 2n + 3$ for all $n = 0, 1, 2, \dots$?

A. $\frac{3}{(1-x)^2}$

B. $\frac{3x}{(1-x)^2}$

C. $\frac{2-x}{(1-x)^2}$

D. $\frac{3-x}{(1-x)^2}$

gatecse-2018 generating-functions normal combinatory one-mark

Answer key

1.4.6 Generating Functions: GATE CSE 2022 | Question: 26



Which one of the following is the closed form for the generating function of the sequence $\{a_n\}_{n \geq 0}$ defined below?

$$a_n = \begin{cases} n+1, & n \text{ is odd} \\ 1, & \text{otherwise} \end{cases}$$

A. $\frac{x(1+x^2)}{(1-x^2)^2} + \frac{1}{1-x}$

B. $\frac{x(3-x^2)}{(1-x^2)^2} + \frac{1}{1-x}$

C. $\frac{2x}{(1-x^2)^2} + \frac{1}{1-x}$

D. $\frac{x}{(1-x^2)^2} + \frac{1}{1-x}$

gatecse-2022 combinatory generating-functions two-marks

Answer key

1.5 Modular Arithmetic (2)

1.5.1 Modular Arithmetic: GATE CSE 2016 | Set 2 | Question: 29



The value of the expression $13^{99} \pmod{17}$ in the range 0 to 16, is _____.

gatecse-2016-set2 modular-arithmetic normal numerical-answers

Answer key

1.5.2 Modular Arithmetic: GATE CSE 2019 | Question: 21



The value of $3^{51} \pmod{5}$ is _____.

gatecse-2019 numerical-answers combinatory modular-arithmetic one-mark

Answer key

1.6 Pigeonhole Principle (2)

1.6.1 Pigeonhole Principle: GATE CSE 2000 | Question: 1.1



The minimum number of cards to be dealt from an arbitrarily shuffled deck of 52 cards to guarantee that three cards are from same suit is

A. 3

B. 8

C. 9

D. 12

gatecse-2000 easy pigeonhole-principle combinatory

Answer key

1.6.2 Pigeonhole Principle: GATE CSE 2005 | Question: 44



What is the minimum number of ordered pairs of non-negative numbers that should be chosen to ensure that there are two pairs (a, b) and (c, d) in the chosen set such that, $a \equiv c \pmod{3}$ and $b \equiv d \pmod{5}$

A. 4

B. 6

C. 16

D. 24

gatecse-2005 set-theory&algebra normal pigeonhole-principle

Answer key

1.7 Recurrence Relation (7)

1.7.1 Recurrence Relation: GATE CSE 1996 | Question: 9



The Fibonacci sequence $\{f_1, f_2, f_3 \dots f_n\}$ is defined by the following recurrence:

$$f_{n+2} = f_{n+1} + f_n, n \geq 1; f_2 = 1 : f_1 = 1$$

Prove by induction that every third element of the sequence is even.

gate1996 recurrence-relation proof descriptive

Answer key

1.7.2 Recurrence Relation: GATE CSE 2016 | Set 1 | Question: 2



Let a_n be the number of n -bit strings that do NOT contain two consecutive 1's. Which one of the following is the recurrence relation for a_n ?

A. $a_n = a_{n-1} + 2a_{n-2}$

B. $a_n = a_{n-1} + a_{n-2}$

C. $a_n = 2a_{n-1} + a_{n-2}$

D. $a_n = 2a_{n-1} + 2a_{n-2}$

gatecse-2016-set1 combinatory recurrence-relation easy

Answer key

1.7.3 Recurrence Relation: GATE CSE 2016 | Set 1 | Question: 27



Consider the recurrence relation $a_1 = 8, a_n = 6n^2 + 2n + a_{n-1}$. Let $a_{99} = K \times 10^4$. The value of K is _____.

gatecse-2016-set1 combinatory recurrence-relation normal numerical-answers

Answer key

1.7.4 Recurrence Relation: GATE CSE 2022 | Question: 41



Consider the following recurrence:

$$\begin{aligned} f(1) &= 1; \\ f(2n) &= 2f(n) - 1, \quad \text{for } n \geq 1; \\ f(2n+1) &= 2f(n) + 1, \quad \text{for } n \geq 1. \end{aligned}$$

Then, which of the following statements is/are TRUE?

A. $f(2^n - 1) = 2^n - 1$

B. $f(2^n) = 1$

C. $f(5 \cdot 2^n) = 2^{n+1} + 1$

D. $f(2^n + 1) = 2^n + 1$

gatecse-2022 combinatory recurrence-relation multiple-selects two-marks

Answer key

1.7.5 Recurrence Relation: GATE CSE 2023 | Question: 5



The Lucas sequence L_n is defined by the recurrence relation:

$$L_n = L_{n-1} + L_{n-2}, \quad \text{for } n \geq 3,$$

with $L_1 = 1$ and $L_2 = 3$.

Which one of the options given is TRUE?

A. $L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$

B. $L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{3}\right)^n$

C. $L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{3}\right)^n$

D. $L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n$

gatecse-2023 combinatory recurrence-relation one-mark

Answer key

1.7.6 Recurrence Relation: GATE IT 2004 | Question: 34



Let $H_1, H_2, H_3, \dots$ be harmonic numbers. Then, for $n \in \mathbb{Z}^+$, $\sum_{j=1}^n H_j$ can be expressed as

A. $nH_{n+1} - (n+1)$
 C. $nH_n - n$

B. $(n+1)H_n - n$
 D. $(n+1)H_{n+1} - (n+1)$

gateit-2004 recurrence-relation combinatory normal

Answer key

1.7.7 Recurrence Relation: GATE IT 2007 | Question: 76



Consider the sequence $\langle x_n \rangle$, $n \geq 0$ defined by the recurrence relation $x_{n+1} = c \cdot x_n^2 - 2$, where $c > 0$.

Suppose there exists a **non-empty, open** interval (a, b) such that for all x_0 satisfying $a < x_0 < b$, the sequence converges to a limit. The sequence converges to the value?

A. $\frac{1+\sqrt{1+8c}}{2c}$

B. $\frac{1-\sqrt{1+8c}}{2c}$

C. 2

D. $\frac{2}{2c-1}$

gateit-2007 combinatory normal recurrence-relation

Answer key

1.8 Summation (4)

1.8.1 Summation: GATE CSE 1989 | Question: 4-i



How many substrings (of all lengths inclusive) can be formed from a character string of length n ? Assume all characters to be distinct, prove your answer.

gate1989 descriptive combinatory normal proof counting summation

Answer key

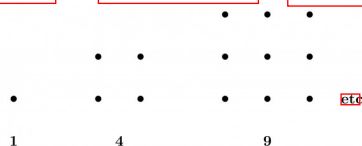
1.8.2 Summation: GATE CSE 1994 | Question: 15



Use the patterns given to prove that

A. $\sum_{i=0}^{n-1} (2i+1) = n^2$

(You are not permitted to employ induction)



B. Use the result obtained in (A) to prove that $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

gate1994 combinatory proof summation descriptive

Answer key

1.8.3 Summation: GATE CSE 2008 | Question: 24



Let $P = \sum_{i \text{ odd}}^{1 \leq i \leq 2k} i$ and $Q = \sum_{i \text{ even}}^{1 \leq i \leq 2k} i$, where k is a positive integer. Then

A. $P = Q - k$

B. $P = Q + k$

C. $P = Q$

D. $P = Q + 2k$

gatecse-2008 combinatory easy summation

Answer key



$$\sum_{x=1}^{99} \frac{1}{x(x+1)} = \text{_____}.$$

gatecse-2015-set1

combinatory

normal

numerical-answers

summation

Answer key

Answer Keys

1.1.1	D	1.1.2	N/A	1.1.3	N/A	1.1.4	B	1.1.5	7
1.1.6	A	1.2.1	A	1.2.2	N/A	1.2.3	D	1.2.4	D
1.2.5	B	1.2.6	C	1.2.7	B	1.2.8	C	1.2.9	A
1.2.10	D	1.2.11	89	1.2.12	15	1.2.13	C	1.2.14	12
1.2.15	195:195	1.2.16	A	1.2.17	B	1.2.18	A	1.3.1	A;C
1.3.2	N/A	1.3.3	D	1.3.4	65 : 65	1.3.5	59049 : 59049	1.3.6	D
1.4.1	N/A	1.4.2	B	1.4.3	10	1.4.4	15	1.4.5	D
1.4.6	A	1.5.1	4	1.5.2	2	1.6.1	C	1.6.2	C
1.7.1	N/A	1.7.2	B	1.7.3	197.9 : 198.1	1.7.4	A;B;C	1.7.5	A
1.7.6	B	1.7.7	B	1.8.1	N/A	1.8.2	N/A	1.8.3	A
1.8.4	0.99								



Syllabus: Connectivity, Matching, Coloring.

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	0	0	0	0	1	1	0	1	1	0	0
2 Marks Count	3	1	0	0	1	2	1	3	1	0	0
Total Marks	6	2	0	0	3	5	2	7	3	0	0

Welcome to the Graph Theory chapter, a cornerstone of Discrete Mathematics and a frequently tested area in the GATE Computer Science examination. Graph Theory provides a powerful framework for modeling relationships and processes across various domains, from social networks and transportation systems to computer networks and algorithm design. For GATE CS, this subject typically carries a weightage of 5-8 marks, often integrated with Data Structures and Algorithms. Questions range from direct application of definitions and formulas, analysis of graph properties, tracing graph algorithms, to problem-solving scenarios involving connectivity, coloring, and planarity. A strong grasp of graph theory fundamentals is crucial not only for scoring well but also for understanding advanced topics in computer science.

Topic-wise Key Concepts

Counting

Counting in graph theory involves determining the number of possible graphs, paths, cycles, or specific structures within graphs. It often relies on fundamental combinatorial principles.

- **Definition and Core Idea:** This topic focuses on enumerating different types of graphs or substructures within them, using principles of permutations, combinations, and other combinatorial techniques. It's about understanding how many ways a graph can be formed or how many specific patterns exist.

Important Formulas, Theorems, and Results:

1. **Number of simple labeled graphs with n vertices:**

$$2^{\binom{n}{2}} = 2^{n(n-1)/2}$$

This is because for each pair of n vertices, there can either be an edge or not.

2. **Number of simple directed labeled graphs with n vertices:**

$$2^{n(n-1)}$$

For each ordered pair of distinct vertices (u, v) , there can be an edge from u to v , or not.

3. **Number of spanning trees in a complete graph K_n (Cayley's Formula):**

$$n^{n-2}$$

This formula is critical for counting trees on a given set of labeled vertices.

4. **Number of spanning trees in a general graph (Matrix Tree Theorem):** This theorem uses the Laplacian matrix of a graph, but its direct application is usually beyond GATE scope; Cayley's formula is more common.
5. **Number of paths of length k between two vertices in an adjacency matrix:** The (i, j) -th entry of A^k (where A is the adjacency matrix) gives the number of paths of length k from vertex i to vertex j .

Key Properties and Identities:

- Understanding the difference between labeled and unlabeled graphs is crucial. GATE questions usually imply labeled graphs unless specified.
- Basic combinatorial identities like $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ and $P(n, k) = \frac{n!}{(n-k)!}$ are fundamental.
- The Inclusion-Exclusion Principle can be used for more complex counting problems, especially when dealing with properties that overlap.

Common Pitfalls or Tricky Points:

- **Overcounting or Undercounting:** Ensure each distinct configuration is counted exactly once.
- **Distinguishing Labeled vs. Unlabeled:** Most GATE questions assume labeled vertices unless stated otherwise, which simplifies counting. Unlabeled graph counting is significantly harder.
- **Specific Graph Types:** Be careful when counting paths/cycles in specific graph types like trees, complete graphs, or bipartite graphs.

- **Standard Problem-Solving Techniques or Shortcuts:**

- **Direct Application of Formulas:** For simple graphs, complete graphs, and spanning trees, direct formula application is common.
- **Combinatorial Arguments:** Break down the problem into smaller, manageable choices and multiply/add the possibilities.
- **Recurrence Relations:** For counting paths or specific structures, sometimes a recurrence relation can be formulated and solved.

Degree of Graph

The degree of a vertex is a fundamental property that quantifies its connectivity within a graph. It's essential for understanding graph structure and validating basic graph properties.

- **Definition and Core Idea:** The degree of a vertex v , denoted $\deg(v)$ or $d(v)$, is the number of edges incident to it. For directed graphs, we distinguish between in-degree (edges pointing to v) and out-degree (edges pointing from v).

- **Important Formulas, Theorems, and Results:**

1. **Handshaking Lemma (for undirected graphs):** The sum of the degrees of all vertices in any undirected graph is equal to twice the number of edges.

$$\sum_{v \in V} \deg(v) = 2|E|$$

This is a cornerstone theorem in graph theory.

2. **For directed graphs:** The sum of in-degrees equals the sum of out-degrees, and both are equal to the total number of edges.

$$\sum_{v \in V} \text{in-deg}(v) = \sum_{v \in V} \text{out-deg}(v) = |E|$$

3. **Number of odd-degree vertices:** In any undirected graph, the number of vertices with odd degree must be even. This is a direct consequence of the Handshaking Lemma.
4. **Maximum degree $\Delta(G)$ and Minimum degree $\delta(G)$:** These denote the maximum and minimum degrees among all vertices in graph G , respectively.

- **Key Properties and Identities:**

- The Handshaking Lemma is a powerful tool for checking the validity of a degree sequence or finding the number of edges.
- A graph cannot exist if its degree sequence violates the Handshaking Lemma or has an odd number of odd-degree vertices.
- For a simple graph with n vertices, the maximum degree of any vertex is $n - 1$.

- **Common Pitfalls or Tricky Points:**

- **Self-loops:** In some contexts, a self-loop at a vertex contributes 2 to its degree. GATE questions usually specify if self-loops are allowed or assume simple graphs (no self-loops, no multiple edges).
- **Multiple Edges:** If multiple edges are allowed between two vertices, each edge contributes to the degree.
- **Directed vs. Undirected:** Always be clear whether the graph is directed or undirected when calculating degrees.

- **Standard Problem-Solving Techniques or Shortcuts:**

- **Direct Application of Handshaking Lemma:** Often used to find the number of edges given a degree sequence, or to determine if a given degree sequence is possible.
- **Parity Check:** Quickly identify impossible degree sequences by checking the number of odd-degree vertices.

Graph Algorithms

Graph algorithms are systematic procedures for solving problems on graphs, such as finding paths, spanning trees, or optimal routes. They are central to computer science applications.

- **Definition and Core Idea:** These are algorithms designed to traverse, search, analyze, and optimize properties of graphs. They form the backbone of many real-world applications, from network routing to social network analysis.

- **Important Algorithms, Formulas, and Complexities:**

1. **Breadth-First Search (BFS):** Explores a graph layer by layer.
 - **Purpose:** Find shortest path in unweighted graphs, find connected components.
 - **Time Complexity:** $O(V + E)$ for adjacency list, $O(V^2)$ for adjacency matrix.
2. **Depth-First Search (DFS):** Explores as far as possible along each branch before backtracking.
 - **Purpose:** Find connected components, topological sort, detect cycles, find bridges/articulation points.

- **Time Complexity:** $O(V + E)$ for adjacency list, $O(V^2)$ for adjacency matrix.
- 3. **Dijkstra's Algorithm:** Finds the shortest paths from a single source vertex to all other vertices in a graph with non-negative edge weights.
 - **Purpose:** Single-source shortest path.
 - **Time Complexity:** $O(E \log V)$ with a Fibonacci heap, $O(E + V \log V)$ with a binary heap, $O(V^2)$ with adjacency matrix.
- 4. **Prim's Algorithm:** Finds a Minimum Spanning Tree (MST) in a weighted, undirected graph.
 - **Purpose:** MST.
 - **Time Complexity:** $O(E \log V)$ with a binary heap, $O(V^2)$ with adjacency matrix.
- 5. **Kruskal's Algorithm:** Finds a Minimum Spanning Tree (MST) in a weighted, undirected graph.
 - **Purpose:** MST.
 - **Time Complexity:** $O(E \log E)$ or $O(E \log V)$ using a Disjoint Set Union (DSU) data structure.
- 6. **Bellman-Ford Algorithm:** Finds the shortest paths from a single source vertex to all other vertices, even with negative edge weights. Detects negative cycles.
 - **Purpose:** Single-source shortest path with negative weights.
 - **Time Complexity:** $O(VE)$.
- 7. **Floyd-Warshall Algorithm:** Finds all-pairs shortest paths in a weighted graph, can handle negative weights but not negative cycles.
 - **Purpose:** All-pairs shortest path.
 - **Time Complexity:** $O(V^3)$.
- 8. **Topological Sort:** Linear ordering of vertices such that for every directed edge (u, v) , u comes before v . Applicable only to Directed Acyclic Graphs (DAGs).
 - **Purpose:** Ordering tasks with dependencies.
 - **Time Complexity:** $O(V + E)$ (using DFS or Kahn's algorithm).
- 9. **Ford-Fulkerson Algorithm (and Edmonds-Karp):** Finds the maximum flow in a flow network.
 - **Purpose:** Maximum flow.
 - **Time Complexity:** Ford-Fulkerson is not polynomial in general; Edmonds-Karp is $O(VE^2)$.
- **Key Properties and Identities:**
 - **Greedy Algorithms:** Prim's, Kruskal's, Dijkstra's are greedy algorithms that make locally optimal choices.
 - **Dynamic Programming:** Floyd-Warshall and Bellman-Ford use dynamic programming principles.
 - **Connectivity:** BFS/DFS can determine if a graph is connected or find its connected components.
 - **Cycles:** DFS can detect cycles in both directed and undirected graphs.
- **Common Pitfalls or Tricky Points:**
 - **Negative Edge Weights:** Dijkstra's fails with negative edge weights; use Bellman-Ford or Floyd-Warshall.
 - **Disconnected Graphs:** Algorithms like Dijkstra's might not reach all vertices if the graph is disconnected from the source.
 - **Choosing the Right Algorithm:** Understand the constraints (weighted/unweighted, directed/undirected, negative weights, single-source/all-pairs) to select the appropriate algorithm.
 - **Adjacency List vs. Matrix:** Be aware of how the graph representation affects time complexity.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - **Tracing:** Practice tracing algorithms step-by-step on small graphs to understand their mechanics.
 - **Recognizing Patterns:** Identify if a problem can be mapped to a standard graph algorithm (e.g., "minimum cost to connect all cities" suggests MST).
 - **Complexity Analysis:** Be able to derive or recall the time and space complexity for common algorithms.

Graph Coloring

Graph coloring assigns labels (colors) to graph elements subject to certain constraints, often used to model scheduling or resource allocation problems.

- **Definition and Core Idea:** Vertex coloring assigns a color to each vertex such that no two adjacent vertices have the same color. The minimum number of colors required for a graph G is called its chromatic number, denoted $\chi(G)$.
- **Important Formulas, Theorems, and Results:**
 1. **Chromatic Number $\chi(G)$:** The smallest integer k such that G is k -colorable.
 2. **Lower Bound:** For any graph G , $\chi(G) \geq \omega(G)$, where $\omega(G)$ is the clique number (size of the largest clique).
 3. **Upper Bound (Brooks' Theorem):** For a connected graph G that is neither a complete graph nor an odd cycle, $\chi(G) \leq \Delta(G)$, where $\Delta(G)$ is the maximum degree. In general, $\chi(G) \leq \Delta(G) + 1$.
 4. **Bipartite Graphs:** A graph is bipartite if and only if it contains no odd-length cycles. If a graph is bipartite and has at least one edge, then $\chi(G) = 2$.

5. **Planar Graphs (Four Color Theorem):** Every planar graph is 4-colorable. Thus, for any planar graph G , $\chi(G) \leq 4$.
6. **Chromatic Polynomial $P(G, k)$:** A polynomial whose value at integer k is the number of ways to color G using at most k colors. For a graph with n vertices, $P(G, k)$ is a polynomial in k of degree n .
- **Key Properties and Identities:**
 - A graph is 2-colorable if and only if it is bipartite.
 - If a graph contains a clique of size k , then at least k colors are needed.
 - The chromatic number of a complete graph K_n is n .
 - The chromatic number of a cycle graph C_n is 2 if n is even, and 3 if n is odd.
- **Common Pitfalls or Tricky Points:**
 - **Minimum vs. Any Coloring:** The goal is to find the minimum number of colors, not just any valid coloring.
 - **Identifying Cliques:** Finding the largest clique ($\omega(G)$) can be hard, but it provides a lower bound for $\chi(G)$.
 - **Edge Coloring ($\chi'(G)$):** Similar concept but for edges (adjacent edges have different colors). Vizing's Theorem states $\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$. GATE usually focuses on vertex coloring.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - **Greedy Coloring:** Assign colors iteratively to vertices in some order (e.g., decreasing degree). While not always optimal, it's a common heuristic.
 - **Identify Cliques and Odd Cycles:** These structures directly give lower bounds for the chromatic number.
 - **Trial and Error/Backtracking:** For small graphs, try to color with 2, then 3, etc., until a valid coloring is found.

Graph Connectivity

Graph connectivity measures how robust a graph is to the removal of vertices or edges. It's crucial for network reliability and fault tolerance.

- **Definition and Core Idea:** Connectivity refers to the property of a graph being "connected," meaning there's a path between any two vertices. More generally, it quantifies how many vertices or edges must be removed to disconnect the graph.
- **Important Formulas, Theorems, and Results:**
 1. **Connected Graph:** A graph is connected if there is a path between every pair of distinct vertices.
 2. **Connected Components:** Maximal connected subgraphs.
 3. **Cut Vertex (Articulation Point):** A vertex whose removal increases the number of connected components.
 4. **Cut Edge (Bridge):** An edge whose removal increases the number of connected components.
 5. **Vertex Connectivity $\kappa(G)$:** The minimum number of vertices whose removal disconnects G or reduces it to a trivial graph. A graph is k -connected if $\kappa(G) \geq k$.
 6. **Edge Connectivity $\lambda(G)$:** The minimum number of edges whose removal disconnects G . A graph is k -edge-connected if $\lambda(G) \geq k$.
 7. **Relationship:** For any graph G , $\kappa(G) \leq \lambda(G) \leq \delta(G)$, where $\delta(G)$ is the minimum degree of G .
 8. **Menger's Theorem:** For any two non-adjacent vertices u and v in a graph G , the maximum number of vertex-disjoint paths between u and v is equal to the minimum number of vertices whose removal separates u and v . (Similar theorem for edge-disjoint paths and edge cuts).
- **Key Properties and Identities:**
 - A tree with more than one vertex has $\kappa(G) = 1$ and $\lambda(G) = 1$.
 - A complete graph K_n has $\kappa(K_n) = n - 1$ and $\lambda(K_n) = n - 1$.
 - If a graph has a bridge, its edge connectivity is 1. If it has an articulation point, its vertex connectivity is 1 (unless it's K_2).
- **Common Pitfalls or Tricky Points:**
 - **Trivial Graph:** For a complete graph K_n , removing $n - 1$ vertices disconnects it into a trivial graph (single vertex), so $\kappa(K_n) = n - 1$.
 - **Distinguishing Vertex vs. Edge Connectivity:** Carefully read whether the question asks for vertex or edge removal.
 - **Disconnected Graphs:** For a disconnected graph, $\kappa(G) = 0$ and $\lambda(G) = 0$.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - **BFS/DFS:** Can be used to find connected components, bridges, and articulation points.
 - **Trial and Error:** For small graphs, try removing vertices/edges to find the minimum set that disconnects the graph.
 - **Apply $\kappa(G) \leq \lambda(G) \leq \delta(G)$:** This inequality provides quick bounds.

Graph Isomorphism

Graph isomorphism determines if two graphs are structurally identical, regardless of their vertex labeling or drawing style. It's a fundamental concept for comparing graphs.

- **Definition and Core Idea:** Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there exists a bijective function $f: V_1 \rightarrow V_2$ such that for any pair of vertices $u, v \in V_1$, $\{u, v\} \in E_1$ if and only if $\{f(u), f(v)\} \in E_2$. Essentially, they have the same structure.
- **Important Properties (Invariants):**
 1. **Number of Vertices:** $|V_1| = |V_2|$
 2. **Number of Edges:** $|E_1| = |E_2|$
 3. **Degree Sequence:** Both graphs must have the same degree sequence (multiset of degrees).
 4. **Number of Connected Components:** Must be the same.
 5. **Cycle Lengths:** The number of cycles of each specific length must be the same.
 6. **Connectivity:** Same vertex and edge connectivity.
 7. **Chromatic Number:** Must be the same.
 8. **Presence of Subgraphs:** If G_1 contains a subgraph H , then G_2 must also contain H .
- **Key Properties and Identities:**
 - ◻ If any of the invariants differ, the graphs are definitely NOT isomorphic.
 - ◻ If all invariants are the same, the graphs MIGHT be isomorphic, but it's not a guarantee (e.g., regular graphs can have the same degree sequence but be non-isomorphic).
 - ◻ The Graph Isomorphism Problem (determining if two graphs are isomorphic) is in NP, but its exact complexity class is unknown (neither P nor NP-complete is proven). For GATE, problems are usually small enough to solve by inspection or invariant checking.
- **Common Pitfalls or Tricky Points:**
 - ◻ **Proving Non-Isomorphism:** This is generally easier; just find one invariant that differs.
 - ◻ **Proving Isomorphism:** This requires finding an explicit bijective mapping f , which can be challenging. Often, it involves matching vertices with identical properties (e.g., same degree, same neighborhood structure).
 - ◻ **Visual Deception:** Graphs drawn differently can be isomorphic, and graphs drawn similarly can be non-isomorphic.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - ◻ **Check Invariants Systematically:** Start with simple invariants like number of vertices, edges, and degree sequence. If any differ, stop.
 - ◻ **Match Vertices by Properties:** Look for unique vertices (e.g., a vertex with degree 1, or a vertex part of a unique cycle structure) and try to map them.
 - ◻ **Adjacency Matrix Comparison:** If two graphs are isomorphic, their adjacency matrices can be made identical by permuting rows and columns. This is computationally intensive for humans but conceptually useful.

Graph Matching

Graph matching is about finding a set of edges that do not share any common vertices, often used in assignment problems or resource allocation.

- **Definition and Core Idea:** A matching M in a graph G is a set of edges such that no two edges in M share a common vertex. A vertex is *matched* (or *covered*) if it is an endpoint of an edge in M ; otherwise, it is *unmatched*.
- **Important Formulas, Theorems, and Results:**
 1. **Maximum Matching:** A matching with the largest possible number of edges.
 2. **Perfect Matching:** A matching that covers all vertices of the graph. A perfect matching can only exist if $|V|$ is even.
 3. **Augmenting Path:** A path that starts and ends at unmatched vertices, and whose edges alternate between being in the current matching and not in the current matching. An augmenting path can be used to increase the size of a matching.
 4. **Hall's Marriage Theorem (for Bipartite Graphs):** A bipartite graph $G = (U \cup V, E)$ has a matching that covers all vertices in U if and only if for every subset $A \subseteq U$, $|N(A)| \geq |A|$, where $N(A)$ is the set of neighbors of vertices in A .
 5. **Tutte's Theorem (for General Graphs):** A graph G has a perfect matching if and only if for every subset $S \subseteq V$, the number of odd components in $G - S$ is less than or equal to $|S|$. (This is more complex and less common in GATE than Hall's).
- **Key Properties and Identities:**
 - ◻ The size of a maximum matching in a bipartite graph can be found using network flow algorithms (e.g., by converting to a max-flow problem).
 - ◻ Every perfect matching is a maximum matching, but not vice-versa.
 - ◻ A graph can have multiple maximum matchings.

- **Common Pitfalls or Tricky Points:**

- **Confusing Matching with Vertex Cover or Edge Cover:** These are related but distinct concepts. A vertex cover is a set of vertices that touch all edges. An edge cover is a set of edges that touch all vertices (only exists if no isolated vertices).
- **General vs. Bipartite Graphs:** Finding maximum matchings in general graphs is more complex (e.g., Edmonds' blossom algorithm) than in bipartite graphs. GATE usually focuses on bipartite matching or simpler cases.
- **Existence of Perfect Matching:** A perfect matching requires an even number of vertices.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - **Greedy Approach:** While not always optimal for maximum matching, a greedy approach (e.g., pick an edge, remove its endpoints, repeat) can give a good starting point.
 - **Augmenting Paths:** For bipartite graphs, the concept of augmenting paths is key. Start with an empty matching, find an augmenting path, update the matching, and repeat until no more augmenting paths exist. This is the basis of the Hopcroft-Karp algorithm.
 - **Applying Hall's Theorem:** For bipartite graphs, use Hall's condition to check for the existence of a matching covering one side.

Graph Planarity

Graph planarity deals with whether a graph can be drawn on a plane without any edges crossing. It has applications in circuit design and network visualization.

- **Definition and Core Idea:** A graph is planar if it can be drawn in the plane without any edges crossing each other (except at vertices). Such a drawing is called a planar embedding.

- **Important Formulas, Theorems, and Results:**

1. **Euler's Formula for Planar Graphs:** For any connected planar graph with V vertices, E edges, and F faces (regions),

$$V - E + F = 2$$

This formula is fundamental for planar graphs.

2. **Corollaries of Euler's Formula (for simple connected planar graphs with $V > 3$):**

- If G has no cycles of length 3 (i.e., no K_3 subgraph), then $E < 2V - 4$.
- In general, $E < 3V - 6$.
- There is at least one vertex with degree at most 5 ($\delta(G) \leq 5$).

3. **Kuratowski's Theorem:** A graph is planar if and only if it does not contain a subgraph that is a subdivision of K_5 (complete graph on 5 vertices) or $K_{3,3}$ (complete bipartite graph with 3 vertices in each partition).
 - A subdivision of a graph is obtained by replacing edges with paths.

- **Key Properties and Identities:**

- K_5 and $K_{3,3}$ are non-planar.
- Any graph that contains K_5 or $K_{3,3}$ as a minor (or subdivision) is non-planar.
- The dual of a planar graph is also planar.

- **Common Pitfalls or Tricky Points:**

- **Misapplying Euler's Formula:** Remember it's for *connected* planar graphs. For disconnected graphs, $V - E + F = 1 + k$ where k is the number of connected components.
- **Identifying Subdivisions:** Recognizing K_5 or $K_{3,3}$ subdivisions can be tricky. Look for 5 vertices all connected to each other (possibly via paths) or 6 vertices arranged in a bipartite manner.
- **Different Drawings:** A graph might look non-planar in one drawing but be planar in another. The definition requires *existence* of a planar drawing.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - **Check Euler's Formula Corollaries:** If $E > 3V - 6$ (or $E > 2V - 4$ if no triangles), the graph is definitely non-planar. This is a quick check for non-planarity.
 - **Look for K_5 or $K_{3,3}$ Subdivisions:** Try to simplify the graph by removing degree-2 vertices (which don't affect planarity for Kuratowski's theorem) or identifying dense subgraphs.
 - **Redrawing:** For small graphs, try to redraw them to eliminate crossings.

Jaccard Coefficient

The Jaccard Coefficient is a statistical measure used to gauge the similarity between two sets, finding applications in data mining, machine learning, and graph analysis (e.g., comparing neighbor sets).

- **Definition and Core Idea:** The Jaccard Coefficient (or Jaccard Index) measures the similarity between two finite

sets, A and B , as the size of their intersection divided by the size of their union. It quantifies the overlap between the sets.

Important Formulas, Theorems, and Results:

1. Jaccard Coefficient $J(A, B)$:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

2. Jaccard Distance $D_J(A, B)$: Measures dissimilarity.

$$D_J(A, B) = 1 - J(A, B) = \frac{|A \cup B| - |A \cap B|}{|A \cup B|}$$

Key Properties and Identities:

- The Jaccard Coefficient ranges from 0 to 1.
 - $J(A, B) = 1$ if $A = B$ (perfect similarity).
 - $J(A, B) = 0$ if $A \cap B = \emptyset$ (no similarity).
- It is symmetric: $J(A, B) = J(B, A)$.
- It is sensitive to the size of the union; a large union with a small intersection leads to a low coefficient.

Common Pitfalls or Tricky Points:

- Miscalculating Intersection or Union:** Ensure correct set operations, especially when dealing with lists that might contain duplicates (convert to sets first).
- Empty Sets:** If both sets are empty, the Jaccard coefficient is usually defined as 1 (perfect similarity). If one is empty and the other is not, it's 0.
- Application Context:** In graph theory, it's often used to compare the neighborhood sets of two vertices to determine their "structural similarity" or to compare sets of features associated with graph nodes.

Standard Problem-Solving Techniques or Shortcuts:

- Direct Calculation:** Identify the elements in the intersection and union, then apply the formula.
- Venn Diagrams:** For small sets, a Venn diagram can help visualize the intersection and union.

Quick Formula Reference

- Counting Simple Labeled Graphs (n vertices):** $2^{\binom{n}{2}}$
- Counting Simple Directed Labeled Graphs (n vertices):** $2^{n(n-1)}$
- Cayley's Formula (Spanning Trees in K_n):** n^{n-2}
- Handshaking Lemma (Undirected):** $\sum_{v \in V} \deg(v) = 2|E|$
- Handshaking Lemma (Directed):** $\sum_{v \in V} \text{in-deg}(v) = \sum_{v \in V} \text{out-deg}(v) = |E|$
- BFS/DFS Time Complexity:** $O(V + E)$ (adj list), $O(V^2)$ (adj matrix)
- Dijkstra's Time Complexity:** $O(E \log V)$ (binary heap), $O(V^2)$ (adj matrix)
- Prim's/Kruskal's Time Complexity (MST):** $O(E \log V)$ or $O(E \log E)$
- Bellman-Ford Time Complexity:** $O(VE)$
- Floyd-Warshall Time Complexity:** $O(V^3)$
- Topological Sort Time Complexity:** $O(V + E)$
- Chromatic Number Lower Bound:** $\chi(G) \geq \omega(G)$ (clique number)
- Chromatic Number Upper Bound (Brooks' Theorem):** $\chi(G) \leq \Delta(G)$ (for non-complete, non-odd cycle connected graphs), else $\chi(G) < \Delta(G) + 1$
- Bipartite Graph Chromatic Number:** $\chi(G) = 2$ (if connected and has edges)
- Planar Graph Chromatic Number (Four Color Theorem):** $\chi(G) \leq 4$
- Connectivity Relationship:** $\kappa(G) \leq \lambda(G) \leq \delta(G)$
- Euler's Formula (Connected Planar Graph):** $V - E + F = 2$
- Planarity Condition (Simple Connected Planar, $V > 3$):** $E < 3V - 6$
- Planarity Condition (Simple Connected Planar, no K_3 , $V > 3$):** $E \leq 2V - 4$
- Kuratowski's Theorem:** Planar iff no K_5 or $K_{3,3}$ subdivision
- Jaccard Coefficient:** $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$
- Jaccard Distance:** $D_J(A, B) = 1 - J(A, B)$

Important Tips for GATE

- Master Definitions and Basic Properties:** Many GATE questions test fundamental definitions (e.g., what is a bridge, what is a perfect matching) and direct consequences of theorems. Ensure you know what each term means precisely.
- Practice Drawing and Visualizing Graphs:** For problems involving planarity, isomorphism, or connectivity, sketching the graph and trying different layouts can be immensely helpful. Don't rely solely on abstract representations.
- Understand Algorithm Mechanics and Complexity:** Don't just memorize complexities; understand how BFS, DFS, Dijkstra's, Prim's, and Kruskal's work step-by-step. Be prepared to trace them on small examples and identify their time/space complexities for both adjacency list and matrix representations.
- Pay Attention to Graph Types and Constraints:** Is the graph directed or undirected? Weighted or unweighted? Simple or multi-graph? Does it allow self-loops? These details significantly impact which formulas or algorithms apply. For example, Dijkstra's fails with negative edge weights.
- Use Invariants for Isomorphism:** When checking for isomorphism, systematically compare invariants like number of vertices, edges, degree sequence, and cycle lengths. If any invariant differs, the graphs are not isomorphic. This is a powerful shortcut.
- Be Wary of Common Pitfalls:** Remember that Euler's formula applies to connected planar graphs. Dijkstra's doesn't work with negative weights. A greedy coloring isn't always optimal. Keep these exceptions in mind.
- Time Management for Tracing Problems:** For algorithm tracing questions, work carefully and systematically. Use scratch paper to keep track of visited nodes, distances, or parent pointers. Avoid rushing, as a single error can invalidate the entire trace.
- Review Special Graph Types:** Be familiar with properties of common graphs like complete graphs (K_n), complete bipartite graphs ($K_{m,n}$), cycle graphs (C_n), path graphs (P_n), and trees. Their specific properties often appear in questions.

2.1

Counting (3)

2.1.1 Counting: GATE CSE 2001 | Question: 2.15



How many undirected graphs (not necessarily connected) can be constructed out of a given set $V = \{v_1, v_2, \dots, v_n\}$ of n vertices?

A. $\frac{n(n-1)}{2}$

B. 2^n

C. $n!$

D. $2^{\frac{n(n-1)}{2}}$

gatecse-2001 graph-theory normal counting

Answer key

2.1.2 Counting: GATE CSE 2004 | Question: 79



How many graphs on n labeled vertices exist which have at least $\frac{(n^2-3n)}{2}$ edges?

A. $\binom{\frac{n^2-n}{2}}{C_{\frac{n^2-3n}{2}}}$

B. $\sum_{k=0}^{\frac{n^2-3n}{2}} \binom{\frac{n^2-3n}{2}}{C_k} \cdot (n^2-n)$

C. $\binom{\frac{n^2-n}{2}}{C_n}$

D. $\sum_{k=0}^n \binom{\frac{n^2-n}{2}}{C_k}$

gatecse-2004 graph-theory combinatory normal counting

Answer key

2.1.3 Counting: GATE CSE 2012 | Question: 38



Let G be a complete undirected graph on 6 vertices. If vertices of G are labeled, then the number of distinct cycles of length 4 in G is equal to

A. 15

B. 30

C. 90

D. 360

gatecse-2012 graph-theory normal marks-to-all counting

Answer key

2.2

Degree of Graph (13)

2.2.1 Degree of Graph: GATE CSE 1987 | Question: 9c



Show that the number of odd-degree vertices in a finite graph is even.

gate1987 graph-theory degree-of-graph descriptive proof

Answer key

2.2.2 Degree of Graph: GATE CSE 1991 | Question: 16-b



Show that all vertices in an undirected finite graph cannot have distinct degrees, if the graph has at least two vertices.

gate1991 graph-theory degree-of-graph descriptive proof

Answer key

2.2.3 Degree of Graph: GATE CSE 1995 | Question: 24



Prove that in finite graph, the number of vertices of odd degree is always even.

gate1995 graph-theory degree-of-graph proof descriptive

Answer key

2.2.4 Degree of Graph: GATE CSE 2003 | Question: 40



A graph $G = (V, E)$ satisfies $|E| \leq 3|V| - 6$. The min-degree of G is defined as $\min_{v \in V} \{\text{degree}(v)\}$. Therefore, min-degree of G cannot be

- A. 3 B. 4 C. 5 D. 6

gatecse-2003 graph-theory normal degree-of-graph

Answer key

2.2.5 Degree of Graph: GATE CSE 2006 | Question: 71



The 2^n vertices of a graph G corresponds to all subsets of a set of size n , for $n > 6$. Two vertices of G are adjacent if and only if the corresponding sets intersect in exactly two elements. The number of vertices of degree zero in G is:

- A. 1 B. n C. $n + 1$ D. 2^n

gatecse-2006 graph-theory normal degree-of-graph

Answer key

2.2.6 Degree of Graph: GATE CSE 2006 | Question: 72



The 2^n vertices of a graph G corresponds to all subsets of a set of size n , for $n > 6$. Two vertices of G are adjacent if and only if the corresponding sets intersect in exactly two elements.

The maximum degree of a vertex in G is:

- A. $\binom{n}{2} \cdot 2^{\frac{n}{2}}$ B. 2^{n-2} C. $2^{n-3} \times 3$ D. 2^{n-1}

gatecse-2006 graph-theory normal degree-of-graph

Answer key

2.2.7 Degree of Graph: GATE CSE 2009 | Question: 3



Which one of the following is **TRUE** for any simple connected undirected graph with more than 2 vertices?

- A. No two vertices have the same degree.
B. At least two vertices have the same degree.
C. At least three vertices have the same degree.
D. All vertices have the same degree.

gatecse-2009 graph-theory normal degree-of-graph

Answer key

2.2.8 Degree of Graph: GATE CSE 2010 | Question: 1



Let $G = (V, E)$ be a graph. Define $\xi(G) = \sum_d i_d * d$, where i_d is the number of vertices of degree d in G .

If S and T are two different trees with $\xi(S) = \xi(T)$, then

- A. $|S| = 2|T|$
C. $|S| = |T|$

- B. $|S| = |T| - 1$
D. $|S| = |T| + 1$

gatecse-2010 graph-theory normal degree-of-graph

Answer key

2.2.9 Degree of Graph: GATE CSE 2010 | Question: 28



The degree sequence of a simple graph is the sequence of the degrees of the nodes in the graph in decreasing order. Which of the following sequences can not be the degree sequence of any graph?

- I. 7, 6, 5, 4, 4, 3, 2, 1
II. 6, 6, 6, 6, 3, 3, 2, 2
III. 7, 6, 6, 4, 4, 3, 2, 2
IV. 8, 7, 7, 6, 4, 2, 1, 1

A. I and II

B. III and IV

C. IV only

D. II and IV

gatecse-2010 graph-theory degree-of-graph

Answer key

2.2.10 Degree of Graph: GATE CSE 2013 | Question: 25



Which of the following statements is/are TRUE for undirected graphs?

P: Number of odd degree vertices is even.

Q: Sum of degrees of all vertices is even.

A. P only

C. Both P and Q

B. Q only

D. Neither P nor Q

gatecse-2013 graph-theory easy degree-of-graph

Answer key

2.2.11 Degree of Graph: GATE CSE 2014 | Set 1 | Question: 52



An ordered n -tuple $(d_1, d_2, \dots, d_n)$ with $d_1 \geq d_2 \geq \dots \geq d_n$ is called *graphic* if there exists a simple undirected graph with n vertices having degrees $d_1, d_2, \dots, d_n$ respectively. Which one of the following 6-tuples is NOT graphic?

A. (1, 1, 1, 1, 1, 1)

C. (3, 3, 3, 1, 0, 0)

B. (2, 2, 2, 2, 2, 2)

D. (3, 2, 1, 1, 1, 0)

gatecse-2014-set1 graph-theory normal degree-of-graph

Answer key

2.2.12 Degree of Graph: GATE CSE 2017 | Set 2 | Question: 23



G is an undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is _____.

gatecse-2017-set2 graph-theory numerical-answers degree-of-graph

Answer key

2.2.13 Degree of Graph: GATE CSE 2026 | Set 1 | Question: 37



Let $G(V, E)$ be a simple, undirected graph. A vertex cover of G is a subset $V' \subseteq V$ such that for every $(u, v) \in E$, $u \in V'$ or $v \in V'$. Let the size of the smallest vertex cover in G be k . Let S be any vertex cover of size k .

For a vertex $v \in V$, which of the following constraints will always ensure that $v \in S$?

- A. The degree of v is at least $k + 1$
- B. The vertex v is on a path of length $k + 1$
- C. The vertex v is on a cycle of length $k + 1$
- D. The vertex v is a part of a clique of size k

gatecse-2026-set1 graph-theory two-marks degree-of-graph multiple-selects

Answer key

2.3 Graph Algorithms (1)

2.3.1 Graph Algorithms: GATE CSE 2026 | Set 1 | Question: 45



An undirected, unweighted, simple graph $G(V, E)$ is said to be 2-colorable if there exists a function $c : V \rightarrow \{0, 1\}$ such that for every $(u, v) \in E, c(u) \neq c(v)$.

Which of the following statements about 2-colorable graphs is/are true?

- A. If G is 2-colorable, then G may contain cycles of odd length
- B. If G is 2-colorable, then G may contain cycles of even length
- C. An optimal algorithm for testing whether G is 2-colorable runs in time $\Theta(|V| + |E|)$, if G is represented as an adjacency list
- D. An optimal algorithm for testing whether G is 2-colorable runs in time $\Theta(|E| \log |V|)$, if G is represented as an adjacency list

gatecse-2026-set1 graph-theory graph-algorithms two-marks multiple-selects

Answer key

2.4 Graph Coloring (11)

2.4.1 Graph Coloring: GATE CSE 2002 | Question: 1.4



The minimum number of colours required to colour the vertices of a cycle with n nodes in such a way that no two adjacent nodes have the same colour is

- A. 2
- B. 3
- C. 4
- D. $n - 2 \lfloor \frac{n}{2} \rfloor + 2$

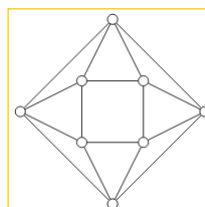
gatecse-2002 graph-theory graph-coloring normal

Answer key

2.4.2 Graph Coloring: GATE CSE 2004 | Question: 77



The minimum number of colours required to colour the following graph, such that no two adjacent vertices are assigned the same color, is



- A. 2
- B. 3
- C. 4
- D. 5

gatecse-2004 graph-theory graph-coloring easy

Answer key

2.4.3 Graph Coloring: GATE CSE 2009 | Question: 2



What is the chromatic number of an n vertex simple connected graph which does not contain any odd length cycle? Assume $n > 2$.

- A. 2 B. 3 C. $n - 1$ D. n

gatecse-2009 graph-theory graph-coloring normal

Answer key

2.4.4 Graph Coloring: GATE CSE 2016 | Set 2 | Question: 03



The minimum number of colours that is sufficient to vertex-colour any planar graph is _____.

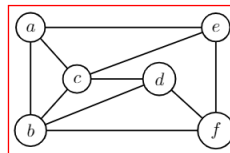
gatecse-2016-set2 graph-theory graph-coloring normal numerical-answers

Answer key

2.4.5 Graph Coloring: GATE CSE 2018 | Question: 18



The chromatic number of the following graph is _____



graph-theory graph-coloring numerical-answers gatecse-2018 one-mark

Answer key

2.4.6 Graph Coloring: GATE CSE 2020 | Question: 52



Graph G is obtained by adding vertex s to $K_{3,4}$ and making s adjacent to every vertex of $K_{3,4}$. The minimum number of colours required to edge-colour G is _____

gatecse-2020 numerical-answers graph-theory graph-coloring two-marks

Answer key

2.4.7 Graph Coloring: GATE CSE 2023 | Question: 45



Let G be a simple, finite, undirected graph with vertex set $\{v_1, \dots, v_n\}$. Let $\Delta(G)$ denote the maximum degree of G and let $\mathbb{N} = \{1, 2, \dots\}$ denote the set of all possible colors. Color the vertices of G using the following greedy strategy: for $i = 1, \dots, n$

$\text{color}(v_i) \leftarrow \min \{j \in \mathbb{N} : \text{no neighbour of } v_i \text{ is colored } j\}$

Which of the following statements is/are TRUE?

- A. This procedure results in a proper vertex coloring of G .
 B. The number of colors used is at most $\Delta(G) + 1$.
 C. The number of colors used is at most $\Delta(G)$.
 D. The number of colors used is equal to the chromatic number of G .

gatecse-2023 graph-theory graph-coloring multiple-selects two-marks

Answer key

2.4.8 Graph Coloring: GATE CSE 2024 | Set 1 | Question: 41



The chromatic number of a graph is the minimum number of colours used in a *proper* colouring of the graph. Let G be any graph with n vertices and chromatic number k . Which of the following statements is/are always TRUE?

- A. G contains a complete subgraph with k vertices

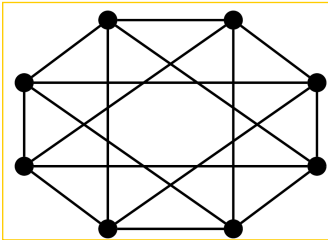
- B. G contains an independent set of size at least n/k
 C. G contains at least $k(k-1)/2$ edges
 D. G contains a vertex of degree at least k

gatecse-2024-set1 multiple-selects graph-theory graph-coloring two-marks

Answer key

2.4.9 Graph Coloring: GATE CSE 2024 | Set 2 | Question: 50

The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. The chromatic number of the following graph is _____.



gatecse-2024-set2 graph-theory numerical-answers graph-coloring two-marks

Answer key

2.4.10 Graph Coloring: GATE IT 2006 | Question: 25

Consider the undirected graph G defined as follows. The vertices of G are bit strings of length n . We have an edge between vertex u and vertex v if and only if u and v differ in exactly one bit position (in other words, v can be obtained from u by flipping a single bit). The ratio of the chromatic number of G to the diameter of G is,

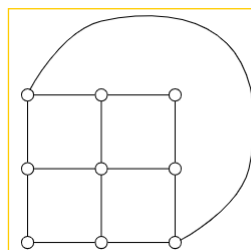
- A. $\frac{1}{(2^n-1)}$ B. $\left(\frac{1}{n}\right)$ C. $\left(\frac{2}{n}\right)$ D. $\left(\frac{3}{n}\right)$

gateit-2006 graph-theory graph-coloring normal

Answer key

2.4.11 Graph Coloring: GATE IT 2008 | Question: 3

What is the chromatic number of the following graph?



- A. 2 B. 3 C. 4 D. 5

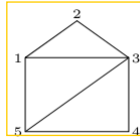
gateit-2008 graph-theory graph-coloring easy

Answer key

2.5 Graph Connectivity (40)

2.5.1 Graph Connectivity: GATE CSE 1987 | Question: 9d





Specify an adjacency-lists representation of the undirected graph given above.

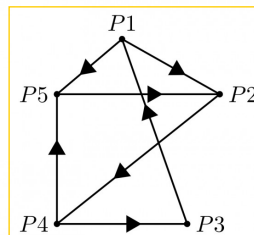
gate1987 graph-theory easy graph-connectivity descriptive

Answer key

2.5.2 Graph Connectivity: GATE CSE 1988 | Question: 2xvi



Write the adjacency matrix representation of the graph given in below figure.



gate1988 descriptive graph-theory graph-connectivity

Answer key

2.5.3 Graph Connectivity: GATE CSE 1990 | Question: 1-viii



A graph which has the same number of edges as its complement must have number of vertices congruent to _____ or _____ modulo 4.

gate1990 graph-theory graph-connectivity fill-in-the-blanks

Answer key

2.5.4 Graph Connectivity: GATE CSE 1991 | Question: 01,xv



The maximum number of possible edges in an undirected graph with n vertices and k components is _____.

gate1991 graph-theory graph-connectivity normal fill-in-the-blanks

Answer key

2.5.5 Graph Connectivity: GATE CSE 1992 | Question: 03,iii



How many edges can there be in a forest with p components having n vertices in all?

gate1992 graph-theory graph-connectivity descriptive

Answer key

2.5.6 Graph Connectivity: GATE CSE 1993 | Question: 8.1



Consider a simple connected graph G with n vertices and n edges ($n > 2$). Then, which of the following statements are true?

- A. G has no cycles
- B. The graph obtained by removing any edge from G is not connected
- C. G has at least one cycle
- D. The graph obtained by removing any two edges from G is not connected
- E. None of the above

gate1993 graph-theory graph-connectivity easy multiple-selects

Answer key

2.5.7 Graph Connectivity: GATE CSE 1994 | Question: 1.6, ISRO2008-29



The number of distinct simple graphs with up to three nodes is

A. 15

B. 10

C. 7

D. 9

gate1994 graph-theory graph-connectivity combinatory normal isro2008 counting

Answer key

2.5.8 Graph Connectivity: GATE CSE 1994 | Question: 2.5



The number of edges in a regular graph of degree d and n vertices is _____

gate1994 graph-theory easy graph-connectivity fill-in-the-blanks

Answer key

2.5.9 Graph Connectivity: GATE CSE 1994 | Question: 24



An independent set in a graph is a subset of vertices such that no two vertices in the subset are connected by an edge. An incomplete scheme for a greedy algorithm to find a maximum independent set in a tree is given below:

```
V: Set of all vertices in the tree;
I :=  $\phi$ 
while  $V \neq \phi$  do
begin
  select a vertex  $u \in V$  such that
   $V := V - \{u\}$ 
  if  $u$  is such that
  then  $I := I \cup \{u\}$ 
end;
Output(I);
```

- A. Complete the algorithm by specifying the property of vertex u in each case.
B. What is the time complexity of the algorithm?

gate1994 graph-theory normal descriptive graph-connectivity

Answer key

2.5.10 Graph Connectivity: GATE CSE 1995 | Question: 1.25



The minimum number of edges in a connected cyclic graph on n vertices is:

A. $n - 1$

B. n

C. $n + 1$

D. None of the above

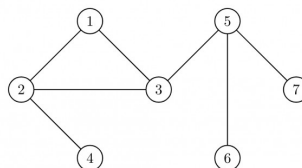
gate1995 graph-theory graph-connectivity easy

Answer key

2.5.11 Graph Connectivity: GATE CSE 1999 | Question: 1.15



The number of articulation points of the following graph is



A. 0

B. 1

C. 2

D. 3

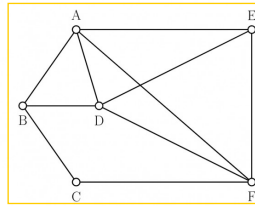
gate1999 graph-theory graph-connectivity normal

Answer key

2.5.12 Graph Connectivity: GATE CSE 1999 | Question: 5



Let G be a connected, undirected graph. A cut in G is a set of edges whose removal results in G being broken into two or more components, which are not connected with each other. The size of a cut is called its cardinality. A min-cut of G is a cut in G of minimum cardinality. Consider the following graph:



- Which of the following sets of edges is a cut?
 - $\{(A, B), (E, F), (B, D), (A, E), (A, D)\}$
 - $\{(B, D), (C, F), (A, B)\}$
- What is cardinality of min-cut in this graph?
- Prove that if a connected undirected graph G with n vertices has a min-cut of cardinality k , then G has at least $\binom{n \times k}{2}$ edges.

gate1999 graph-theory graph-connectivity normal descriptive proof

Answer key

2.5.13 Graph Connectivity: GATE CSE 2002 | Question: 1.25, ISRO2008-30, ISRO2016-6



The maximum number of edges in a n -node undirected graph without self loops is

- n^2
- $\frac{n(n-1)}{2}$
- $n - 1$
- $\frac{(n+1)(n)}{2}$

gatecse-2002 graph-theory easy isro2008 isro2016 graph-connectivity

Answer key

2.5.14 Graph Connectivity: GATE CSE 2003 | Question: 8, ISRO2009-53



Let G be an arbitrary graph with n nodes and k components. If a vertex is removed from G , the number of components in the resultant graph must necessarily lie down between

- k and n
- $k - 1$ and $k + 1$
- $k - 1$ and $n - 1$
- $k + 1$ and $n - k$

gatecse-2003 graph-theory graph-connectivity normal isro2009

Answer key

2.5.15 Graph Connectivity: GATE CSE 2004 | Question: 81



Let $G_1 = (V, E_1)$ and $G_2 = (V, E_2)$ be connected graphs on the same vertex set V with more than two vertices. If $G_1 \cap G_2 = (V, E_1 \cap E_2)$ is not a connected graph, then the graph $G_1 \cup G_2 = (V, E_1 \cup E_2)$

- cannot have a cut vertex
- must have a cycle
- must have a cut-edge (bridge)
- has chromatic number strictly greater than those of G_1 and G_2

gatecse-2004 graph-theory difficult graph-connectivity

Answer key

2.5.16 Graph Connectivity: GATE CSE 2005 | Question: 11



Let G be a simple graph with 20 vertices and 100 edges. The size of the minimum vertex cover of G is 8. Then, the size of the maximum independent set of G is:

- 12
- 8
- less than 8
- more than 12

gatecse-2005 graph-theory normal graph-connectivity

Answer key

2.5.17 Graph Connectivity: GATE CSE 2006 | Question: 73



The 2^n vertices of a graph G corresponds to all subsets of a set of size n , for $n \geq 6$. Two vertices of G are adjacent if and only if the corresponding sets intersect in exactly two elements.

The number of connected components in G is:

A. n

B. $n + 2$

C. $2^{\frac{n}{2}}$

D. $\frac{2^n}{n}$

gatecse-2006 graph-theory normal graph-connectivity

Answer key

2.5.18 Graph Connectivity: GATE CSE 2007 | Question: 23



Which of the following graphs has an Eulerian circuit?

A. Any k -regular graph where k is an even number.

B. A complete graph on 90 vertices.

C. The complement of a cycle on 25 vertices.

D. None of the above

gatecse-2007 graph-theory normal graph-connectivity

Answer key

2.5.19 Graph Connectivity: GATE CSE 2008 | Question: 42



G is a graph on n vertices and $2n - 2$ edges. The edges of G can be partitioned into two edge-disjoint spanning trees. Which of the following is NOT true for G ?

A. For every subset of k vertices, the induced subgraph has at most $2k - 2$ edges.

B. The minimum cut in G has at least 2 edges.

C. There are at least 2 edge-disjoint paths between every pair of vertices.

D. There are at least 2 vertex-disjoint paths between every pair of vertices.

gatecse-2008 graph-connectivity normal

Answer key

2.5.20 Graph Connectivity: GATE CSE 2013 | Question: 26



The line graph $L(G)$ of a simple graph G is defined as follows:

- There is exactly one vertex $v(e)$ in $L(G)$ for each edge e in G .
- For any two edges e and e' in G , $L(G)$ has an edge between $v(e)$ and $v(e')$, if and only if e and e' are incident with the same vertex in G .

Which of the following statements is/are TRUE?

- (P) The line graph of a cycle is a cycle.
- (Q) The line graph of a clique is a clique.
- (R) The line graph of a planar graph is planar.
- (S) The line graph of a tree is a tree.

A. P only

B. P and R only

C. R only

D. P, Q and S only

gatecse-2013 graph-theory normal graph-connectivity

Answer key

2.5.21 Graph Connectivity: GATE CSE 2014 | Set 1 | Question: 51



Consider an undirected graph G where self-loops are not allowed. The vertex set of G is $\{(i, j) \mid 1 \leq i \leq 12, 1 \leq j \leq 12\}$. There is an edge between (a, b) and (c, d) if $|a - c| \leq 1$ and

$|b - d| \leq 1$. The number of edges in this graph is _____.

gatecse-2014-set1 graph-theory numerical-answers normal graph-connectivity

Answer key

2.5.22 Graph Connectivity: GATE CSE 2014 | Set 2 | Question: 3



The maximum number of edges in a bipartite graph on 12 vertices is _____.

gatecse-2014-set2 graph-theory graph-connectivity numerical-answers normal

Answer key

2.5.23 Graph Connectivity: GATE CSE 2014 | Set 3 | Question: 51



If G is the forest with n vertices and k connected components, how many edges does G have?

A. $\lfloor \frac{n}{k} \rfloor$

B. $\lceil \frac{n}{k} \rceil$

C. $n - k$

D. $n - k + 1$

gatecse-2014-set3 graph-theory graph-connectivity normal

Answer key

2.5.24 Graph Connectivity: GATE CSE 2015 | Set 2 | Question: 50



In a connected graph, a bridge is an edge whose removal disconnects the graph. Which one of the following statements is true?

A. A tree has no bridges

B. A bridge cannot be part of a simple cycle

C. Every edge of a clique with size ≥ 3 is a bridge (A clique is any complete subgraph of a graph)

D. A graph with bridges cannot have cycle

gatecse-2015-set2 graph-theory graph-connectivity easy

Answer key

2.5.25 Graph Connectivity: GATE CSE 2019 | Question: 12



Let G be an undirected complete graph on n vertices, where $n > 2$. Then, the number of different Hamiltonian cycles in G is equal to

A. $n!$

B. $(n - 1)!$

C. 1

D. $\frac{(n-1)!}{2}$

gatecse-2019 engineering-mathematics discrete-mathematics graph-theory graph-connectivity one-mark

Answer key

2.5.26 Graph Connectivity: GATE CSE 2019 | Question: 38



Let G be any connected, weighted, undirected graph.

I. G has a unique minimum spanning tree, if no two edges of G have the same weight.

II. G has a unique minimum spanning tree, if, for every cut of G , there is a unique minimum-weight edge crossing the cut.

Which of the following statements is/are TRUE?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

gatecse-2019 engineering-mathematics discrete-mathematics graph-theory graph-connectivity two-marks minimum-spanning-tree algorithms graph-algorithms

Answer key

2.5.27 Graph Connectivity: GATE CSE 2021 | Set 1 | Question: 36



Let $G = (V, E)$ be an undirected unweighted connected graph. The *diameter* of G is defined as:

$$\text{diam}(G) = \max_{u,v \in V} \{\text{the length of shortest path between } u \text{ and } v\}$$

Let M be the adjacency matrix of G .

Define graph G_2 on the same set of vertices with adjacency matrix N , where

$$N_{ij} = \begin{cases} 1 & \text{if } M_{ij} > 0 \text{ or } P_{ij} > 0, \text{ where } P = M^2 \\ 0 & \text{otherwise} \end{cases}$$

Which one of the following statements is true?

- A. $\text{diam}(G_2) < \lceil \text{diam}(G)/2 \rceil$
- B. $\lceil \text{diam}(G)/2 \rceil < \text{diam}(G_2) < \text{diam}(G)$
- C. $\text{diam}(G_2) = \text{diam}(G)$
- D. $\text{diam}(G) < \text{diam}(G_2) \leq 2 \text{diam}(G)$

gatecse-2021-set1 graph-theory graph-connectivity two-marks

Answer key

2.5.28 Graph Connectivity: GATE CSE 2022 | Question: 20



Consider a simple undirected graph of 10 vertices. If the graph is disconnected, then the maximum number of edges it can have is _____.

gatecse-2022 numerical-answers graph-theory graph-connectivity one-mark

Answer key

2.5.29 Graph Connectivity: GATE CSE 2022 | Question: 27



Consider a simple undirected unweighted graph with at least three vertices. If A is the adjacency matrix of the graph, then the number of 3-cycles in the graph is given by the trace of

- A. A^3
- B. A^3 divided by 2
- C. A^3 divided by 3
- D. A^3 divided by 6

gatecse-2022 graph-theory graph-connectivity two-marks

Answer key

2.5.30 Graph Connectivity: GATE CSE 2022 | Question: 42



Which of the properties hold for the adjacency matrix A of a simple undirected unweighted graph having n vertices?

- A. The diagonal entries of A^2 are the degrees of the vertices of the graph.
- B. If the graph is connected, then none of the entries of $A^{n-1} + I_n$ can be zero.
- C. If the sum of all the elements of A is at most $2(n-1)$, then the graph must be acyclic.
- D. If there is at least a 1 in each of A 's rows and columns, then the graph must be connected.

gatecse-2022 graph-theory graph-connectivity multiple-selects two-marks

Answer key

2.5.31 Graph Connectivity: GATE CSE 2024 | Set 1 | Question: 24



The number of spanning trees in a complete graph of 4 vertices labelled A, B, C, and D is _____.

gatecse-2024-set1 numerical-answers graph-theory graph-connectivity one-mark

Answer key

2.5.32 Graph Connectivity: GATE CSE 2024 | Set 2 | Question: 41



Let G be an undirected connected graph in which every edge has a positive integer weight. Suppose that

every spanning tree in G has even weight. Which of the following statements is/are TRUE for every such graph G ?

- A. All edges in G have even weight
- B. All edges in G have even weight OR all edges in G have odd weight
- C. In each cycle C in G , all edges in C have even weight
- D. In each cycle C in G , either all edges in C have even weight OR all edges in C have odd weight

gatecse-2024-set2 graph-theory multiple-selects graph-connectivity two-marks

Answer key

2.5.33 Graph Connectivity: GATE CSE 2024 | Set 2 | Question: 7



Let A be the adjacency matrix of a simple undirected graph G . Suppose A is its own inverse. Which one of the following statements is always TRUE?

- A. G is a cycle
- B. G is a perfect matching
- C. G is a complete graph
- D. There is no such graph G

gatecse-2024-set2 graph-theory graph-connectivity one-mark

Answer key

2.5.34 Graph Connectivity: GATE IT 2004 | Question: 37



What is the number of vertices in an undirected connected graph with 27 edges, 6 vertices of degree 2, 3 vertices of degree 4 and remaining of degree 3?

- A. 10
- B. 11
- C. 18
- D. 19

gateit-2004 graph-theory graph-connectivity normal

Answer key

2.5.35 Graph Connectivity: GATE IT 2004 | Question: 5



What is the maximum number of edges in an acyclic undirected graph with n vertices?

- A. $n - 1$
- B. n
- C. $n + 1$
- D. $2n - 1$

gateit-2004 graph-theory graph-connectivity normal

Answer key

2.5.36 Graph Connectivity: GATE IT 2005 | Question: 56



Let G be a directed graph whose vertex set is the set of numbers from 1 to 100. There is an edge from a vertex i to a vertex j iff either $j = i + 1$ or $j = 3i$. The minimum number of edges in a path in G from vertex 1 to vertex 100 is

- A. 4
- B. 7
- C. 23
- D. 99

gateit-2005 graph-theory graph-connectivity normal

Answer key

2.5.37 Graph Connectivity: GATE IT 2006 | Question: 11



If all the edge weights of an undirected graph are positive, then any subset of edges that connects all the vertices and has minimum total weight is a

- A. Hamiltonian cycle
- B. grid
- C. hypercube
- D. tree

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Answer key